

공학 박사학위 논문

신뢰할 수 있는 인공지능을 위한 주장
정합 보증: 감사 가능한 프라이버시와
안전한 공개 검증

**Claim-Aligned Assurance for Trustworthy AI:
Auditable Privacy and Secure Public Verification**

2026년 8월

서강대학교 가상융합전문대학원

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이 논문을 공학 박사학위 논문으로 제출함

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요약

차분 프라이버시 수치나 워터마크 점수는 보호 대상, 실행 메커니즘, 검증 인터페이스 및 근거가 일치하지 않으면 유효한 보장이 되기 어렵다. 본 논문은 이 간극을 줄이는 주장 정합 보증(Claim-Aligned Assurance)을 제시하고, 감사 가능한 프라이버시와 안전한 공개 검증의 두 축에서 구현한다.

감사 가능한 프라이버시 연구는 완료된 윈도우 기반 DP-SGD 실행의 동결된 근거와 요청 문구를 함께 입력으로 받아, 해당 실행이 실제로 허용하는 프라이버시 단위와 수치의 범위를 판정한다. 실행 단위와 요청 단위가 일치하면 직접 경로를, 등록된 안정성 경계로 변환할 수 있으면 변환 또는 그룹 경로를 사용하며, 근거가 누락되거나 변환 결과가 무의미하면 긍정 문구를 차단한다. 통제된 100개 허용 사례와 100개 대응 비허용 사례에서 전체 판정 튜플이 예상 결과와 일치하였다. WISDM과 UCI HAR 실행은 동일한 수치라도 요청 단위에 따라 허용 범위가 달라짐을 보였고, 5,000명 규모의 Sepsis 자료와 2,779만여 건의 Backblaze 기록은 윈도우 정책에 따라 원시 단위와 생성 레코드 사이의 영향 범위가 크게 달라짐을 확인하였다.

PP-Mark는 생성 문맥에 결속된 잠재 워터마크, 추적 커밋먼트 및 영지식 증명 영수증을 결합한다. 전체 수락은 동일한 문맥과 정확한 이미지에 대한 검출 신호와 유효한 영수증을 모두 요구한다. 10,000개 바인딩 탐색과 화이트박스 최적화는 점수 전용 공격을 드러냈지만, 평가한 위조 조건에서는 전체 수락이 관측되지 않았고 점수 모드는 변환 후 0.74–1.00의 참양성률을 유지했다.

두 연구로부터 주장 명세, 메커니즘 결속, 근거 결속형 판정 및 실패 폐쇄형 보증을 도출한다. 본 논문의 기여는 범용 인증기가 아니라, 긍정적인 프라이버시와 출처 주장을 대상, 메커니즘, 인터페이스 및 근거가 공동으로 지지하는 범위로 제한하는 검토 가능한 보증 구조에 있다.

키워드: 주장 정합 보증, 감사 가능한 프라이버시, 차분 프라이버시, DP-SGD,
안전한 공개 검증, 생성형 AI 출처, 영지식 증명

제 1 장 Introduction

1.1 Research Goals and Significance

Artificial-intelligence systems increasingly communicate trust through compact claims. A trained model may be described as differentially private, a generated image may be labeled as authentic or watermarked, and a public interface may return an acceptance decision. Such statements are useful only when a reader can determine what is protected or verified, which mechanism produced the relevant result, and which evidence supports the wording. A numerical value, a route name, or a detector score is not by itself the guarantee that a public sentence appears to express.

This distinction is especially important when a system crosses several semantic layers. In privacy-preserving learning, raw owners, events, and sequences may be transformed into overlapping windows before a differentially private optimizer operates on them. The reported privacy parameters are meaningful only relative to the privacy unit, neighboring relation, sampling rule, contribution bound, and accounting procedure that generated them [1, 2, 3, 4]. In generative-image provenance, an observable watermark score may survive common transformations, but the score does not alone establish which prompt, seed, model, or embedding trace produced the image. Signed content credentials add an important manifest layer, yet their interpretation still depends on the technical relation actually authenticated by the credential [5, 6].

The central goal of this dissertation is to reduce the gap between such public claims and the system facts that make those claims defensible. It introduces Claim-Aligned Assurance as a design perspective in which a positive trust claim is released only when four elements remain mutually consistent: the intended claim target, the exe-

cuted or registered mechanism, the verification interface, and the available evidence. The framework is instantiated in two research axes: Auditable Privacy, which concerns privacy claims about training, and Secure Public Verification, which concerns independently verifiable provenance for generative-AI content.

The first axis addresses privacy misalignment after a training run has completed. A validator examines frozen evidence together with one requested privacy-unit sentence and determines whether the evidence licenses the native accountant pair, a registered converted consequence, or no positive wording. This focus is practical as well as semantic: a completed model may be reusable for a narrower claim, may support a weaker converted statement, or may require retraining under a mechanism native to the requested unit. The decision cannot repair an incompatible execution, but it can prevent the execution’s isolated accountant output from being silently promoted into a stronger public guarantee.

The second axis addresses a different trust object. Public watermark detectors enable independent screening, but the same public decision surface can be queried or optimized by an adversary [7, 8, 9]. A private detector reduces this exposure at the cost of renewed dependence on a central verifier. This dissertation therefore studies a conjunctive public decision: statistical image-side evidence is combined with a cryptographic receipt bound to the declared generation context and committed embedding trace. The stronger acceptance statement is withheld unless both conditions hold.

The significance of the dissertation lies in this alignment discipline rather than in universal AI certification. The methods do not prove correct execution, universal privacy, or factual truth. They make each positive statement narrower and explicit about its authority. Missing or inconsistent relations therefore withhold positive wording rather than promote an incomplete signal.

1.2 Research Background and Trajectory

The present research problem developed through three preceding studies in anonymous digital systems, decentralized authentication, and context-sensitive AI generation. These studies are not reproduced as independent technical chapters in this dissertation. Their role is to explain how the research focus progressed from interpreting hidden structure, to reducing centralized trust, and then to binding AI generation to domain context. That trajectory motivates the claim–mechanism–evidence question addressed in Chapters 2–3.

1.2.1 Interpreting Hidden Roles in Anonymous Web3 Systems

The first preceding study examined Ethereum, where addresses and transactions are public but the service role of a smart contract and the relationship between accounts are not directly stated. It represented externally owned accounts, contract accounts, and transactions as a heterogeneous graph and used a Graph Attention Network version 2 model for DApp-category classification and account link prediction [10]. The heterogeneous representation supplied role context beyond an address’s isolated balance and transaction values.

The antecedent for this dissertation is the separation between an observable record and an asserted system role. A graph model can infer a role from public traces, but the inference is not an authenticated identity and remained bounded to the evaluated Ethereum data. The scope, model, and evidence determine what may be said. This gap between visible traces, hidden function, and warranted interpretation became the first step toward claim-aligned reasoning.

1.2.2 Reducing Centralized Trust in Metaverse Authentication

The second preceding study moved from interpreting a hidden role to changing who controls a verification decision. Metaverse space access based on centrally managed ranks or passwords leaves the service as both policy authority and point of failure. The proposed approach instead allowed a user to manage an authentication token and used a blockchain smart contract and cosine similarity to evaluate access to a declared virtual space [11]. The study calibrated the similarity threshold and tested feasibility in a metaverse setting.

The resulting lesson is that reducing centralized trust requires alignment among the credential, spatial representation, policy, execution rule, and decision context. A smart contract can make a rule visible and repeatable, but blockchain alone does not make every authentication statement trustworthy. This work therefore contributed a verification structure rather than a universal trust claim.

1.2.3 Preserving Domain Context in LLM Data Augmentation

The third preceding study considered a data problem rather than access control. Sensitive conversational domains can provide few authentic examples while depending on evolving slang, implicit cues, and role-specific behavior. Simple text transformations may increase sample count without preserving this interactional structure. The proposed framework therefore encoded buyer and seller personas, behavioral patterns, domain vocabulary, and dialogue rules in prompts to a large language model [12]. Multi-metric and expert evaluation assessed whether the generated dialogues retained the intended domain characteristics.

The antecedent is that an output-count claim is not a context-fidelity claim. The target and evidence must specify the roles, behavior, language rules, and population being assessed. The study remained limited to a Korean corpus dominated by buyer–

seller dyads, with evolving slang requiring updates. A favorable metric should not be detached from those conditions.

Taken together, the three studies trace a progression in the research question. The Web3 study interpreted hidden roles from public traces; the metaverse study reduced reliance on a centralized authentication decision; and the LLM study bound generation to domain roles and linguistic behavior. The present dissertation advances that progression from inference, decentralization, and context preservation to assurance. It asks when a public privacy or provenance statement is justified by the protected object, operative mechanism, verification interface, and evidence that the system actually exposes.

1.3 Claim-Aligned Assurance for Trustworthy AI

1.3.1 Claim–Mechanism–Evidence Alignment

For this dissertation, a *claim* is a positive, externally interpretable statement about a defined object. A privacy claim names a protected unit, neighboring relation, and privacy parameters. A provenance claim names an image, generation context, and acceptance predicate. A *mechanism* is the set of transformations, randomized procedures, samplers, executors, and cryptographic or statistical operations from which the claim is derived. A *verification interface* defines what a third party submits, observes, and interprets. The *evidence* consists of the artifacts, commitments, runtime records, accountant outputs, proofs, and consistency relations available to support that interpretation.

Let (c), (M), (I), and (E) denote a requested claim, mechanism description, verification interface, and evidence bundle, respectively. Claim-Aligned Assurance is represented as a scoped decision rather than a free-standing trust score:

$$\text{CAA}(c; M, I, E) \in \{\text{Release}(c), \text{Qualify}(c'), \text{Block}\}, \quad c' \prec c. \quad (1.1)$$

The relation $c' \prec c$ means that the evidence supports a narrower or weaker statement than the one requested. For example, frozen DP-SGD evidence may support a generated-window statement but not an owner-level statement, or may support a converted event-level parameter rather than the accountant’s native pair. A watermark score may support statistical screening but not the stronger statement that a declared generation trace has been publicly verified. In both cases, qualification preserves the distinction between what was observed and what was established.

Table 1.1 maps the common alignment elements to the two research axes. It is a dissertation-level synthesis, not an additional empirical result. The domains use different mechanisms and threat models, but each positive statement must close a chain from its declared target to its released decision.

Across these mappings, four design principles recur. First, *claim specification* fixes the target, unit or context, predicate, and scope before a positive label is interpreted. Second, *mechanism binding* connects that specification to the actual route or generation process rather than to a label with a similar name. Third, an *evidence-bound decision* requires the relations needed by the claim, rather than inferring the conclusion from an isolated numerical output. Fourth, *fail-closed assurance* suppresses a positive interpretation when a required relation is absent, inconsistent, unregistered, or vacuous.

표 1.1: Dissertation-wide alignment map for the two research axes

Alignment element	Auditable Privacy	Secure Public Verification
Claim target	Privacy unit, adjacency, and licensed (ε, δ) wording	Exact image, declared generation context, and provenance-acceptance statement
Mechanism binding	Transformation, contribution rule, sampler, clipping/noise geometry, accountant, executor, and schedule	Context-derived payload, latent embedding, trace commitment, image-bound challenge, and proof predicate
Verification interface	Post-execution wording query with an explicit unit and adjacency	Public score mode or conjunctive score-plus-receipt acceptance mode
Evidence	Frozen artifacts, semantic digests, source/runtime relations, recomputed accounting, and registered obligations	Recoverable image signal, declared metadata, Merkle commitment, opened trace tuples, exact image hash, and SP1 receipt
Fail-closed outcome	Block unsupported, unverified, invalid, or vacuous wording	Withhold full acceptance when either the score or receipt condition fails
Principal boundary	Evidence-relative validation is not arbitrary-code proof or execution attestation	Accepted provenance evidence is not semantic truth, human authorship, legal identity, or proof of the complete diffusion execution

The framework treats the verification interface as part of the assurance object. An interface determines which question is being answered and how its output may be interpreted. In the post-execution privacy study, the requested wording is an explicit input because the same frozen run can support different answers for window, event, and owner queries. In PP-Mark, score mode and full acceptance are separate interfaces because they establish different predicates. Interface separation prevents a weaker diagnostic from being silently presented as the stronger claim.

1.3.2 Assurance Boundaries and Fail-Closed Decisions

Claim alignment does not remove the need for a threat model. Internally consistent evidence may still come from a producer that did not execute the reported code, and a proof establishes only the relation encoded by its circuit. Every positive result is therefore relative to explicit premises, evidence authority, and tested scope.

The privacy axis addresses an honest-but-fallible development setting. Digests, independent recomputation, and finite validation predicates expose registered inconsistencies but do not attest hidden execution or arbitrary extensions. The provenance axis considers adversarial signal creation, yet full acceptance remains conditional on its hash, proof-system, key-control, statistical, predicate, and partial-opening assumptions. Its exact-image receipt must be reissued after an edit, and its circuit neither re-executes complete diffusion nor decides factual truth.

Fail-closed behavior marks an evidentiary boundary, not absence of the broader property. A blocked owner claim lacks licensing evidence, and a rejected PP-Mark submission fails the declared score-and-receipt predicate. Neither decision alone proves that the broader privacy or provenance property is false.

1.4 Research Questions and Contributions

The dissertation addresses three research questions.

- RQ1.** Given a completed window-based DP-SGD execution and a requested privacy-unit sentence, which registered evidence obligations and transformations are sufficient to release, qualify, or block that exact wording?
- RQ2.** How can generative-image provenance remain publicly verifiable while requiring more than an optimizable watermark score?
- RQ3.** Which design principles are shared by these privacy and provenance settings, and where must their assurance conclusions explicitly stop?

1.4.1 Auditable Privacy: Post-Execution Claim Validation

Chapter 2 addresses RQ1 after training. Given a frozen evidence bundle and wording query, it reopens the raw-to-generated mapping, checks mechanism and accountant relations, and returns a direct, converted, grouped, underspecified, or blocked path. Its controlled conformance study, fixed information-necessity witnesses, protocol-frozen activity-data executions, and Sepsis and Backblaze mapping diagnostics test the decision rather than propose a new accountant. The resulting Auditable Privacy contribution is an inspectable chain from a requested sentence to the observable premises that license or block it. It remains evidence-relative and is not promoted into end-to-end execution attestation or a general proof of differential privacy.

1.4.2 Secure Public Verification: Detection and Verifiable Provenance

Chapter 3 addresses RQ2 through PP-Mark. A secret-keyed generation binding determines the latent payload and sampled trace, which is Merkle-committed and par-

tially opened through an image-bound SP1 proof. PP-Mark(score) remains a statistical screen; PP-Mark(accept) requires both the calibrated score and a valid receipt for the same context and exact image. Forgery, removal, transformation, cost, quality, and bounded model-generalization experiments evaluate the two interfaces without equating either with semantic truth.

RQ3 is answered by the shared framework rather than by treating the two core studies as one algorithm. Across them, the dissertation contributes:

- a common vocabulary for specifying a trust claim by its target, mechanism, verification interface, evidence, and assurance boundary;
- a post-execution, query-dependent validator that restricts privacy wording to what a frozen evidence chain licenses;
- a public provenance construction whose stronger decision joins a context-bound statistical signal with a publicly verifiable cryptographic receipt; and
- a cross-domain synthesis of claim specification, mechanism binding, evidence-bound decision, and fail-closed behavior, together with explicit non-claims for each instantiation.

The integrated contribution is therefore not a single privacy mechanism or a single watermark detector. It is a disciplined way to determine when a specific trust statement remains aligned with the object, process, interface, and evidence that give the statement meaning, demonstrated in two technically distinct AI assurance domains.

1.5 Dissertation Organization

The remainder of the dissertation is organized as follows.

Chapter 2: Post-Execution Auditing of Privacy-Unit Claims. Chapter 2 introduces the background of privacy units and windowed DP-SGD, formalizes query-dependent evidence-bound validation, and develops direct, converted, grouped, and blocked claim paths. It evaluates the contract through a 200-case controlled oracle, fixed necessity witnesses, independently checked WISDM and UCI HAR executions, and Sepsis and Backblaze mapping diagnostics. The chapter also separates bundle consistency from execution attestation and reports the exact authority of each experimental surface.

Chapter 3: Secure Public Verification of Generative-AI Provenance. Chapter 3 turns to the dissertation’s second axis. It describes context-bound latent watermarking, deterministic trace commitment, SP1 proof generation, and the separate score and full-acceptance interfaces of PP-Mark. It then evaluates forgery, removal, regeneration, transformation, efficiency, quality, and bounded model-generalization conditions. The chapter states the cryptographic and statistical premises under which its public provenance conclusion is interpreted.

Chapter 4: Conclusion. Chapter 4 summarizes the two core studies without repeating their independent conclusions. It relates evidence-bound privacy auditing to secure public provenance verification and states the dissertation-wide design principles of Claim-Aligned Assurance. It concludes with limitations, broader impact, and two concrete future validations: comparison of post-hoc conversion with native patient-level DP, and retrieval of PP-Mark proof artifacts through a C2PA-compatible public manifest repository.

This organization preserves the technical independence of the privacy audit and PP-Mark while giving them one dissertation-level argument. The privacy chapter restricts what a completed execution may claim; the provenance chapter changes the

public acceptance predicate so that observable detection and verifiable process evidence must agree. Chapter 4 returns to this common argument and states precisely what the combined evidence does and does not warrant.

제 2 장 Post-Execution Auditing of Privacy-Unit

Claims: Evidence-Bound Validation

2.1 Introduction

Privacy claims attached to trained artificial intelligence systems often compress a multi-part guarantee into a short numerical label. Differential privacy (DP), however, is a property of a randomized mechanism with respect to a specified neighboring relation; it is not a property of an isolated value of ϵ [1]. A meaningful report must therefore identify at least the privacy unit, adjacency relation, mechanism, sampler, and accountant that support the reported parameters. Contemporary implementation and evaluation guidance likewise treats these elements as part of the guarantee rather than as optional metadata [2, 3, 4].

This requirement becomes difficult to maintain in sequential learning pipelines. Differentially private stochastic gradient descent (DP-SGD) samples training records, clips their contributions, adds calibrated noise, and composes the privacy loss across optimization steps [13]. Yet the records consumed by DP-SGD may be generated windows rather than the raw events, patients, users, or sequences named in a deployment report. Overlapping windows, owner-attribution rules, contribution caps, preprocessing, and data-dependent selection can all change how one raw neighboring change propagates to the generated training data. Consequently, an accountant output for one generated-record adjacency cannot be silently restated as protection for another raw unit.

The resulting problem is not merely whether a completed run has an accountant value. It is whether the available evidence for that run supports one particular public

sentence. Consider a frozen windowed DP-SGD execution and three requested claims. The window-level request may retain the accountant’s base pair, an event-level request may require a weaker converted pair, and an owner-level request may have to be blocked when the registered conversion becomes vacuous. Because the evidence is unchanged while the requested privacy unit and adjacency differ, the claim query must be an explicit input to the decision. A query-oblivious report cannot be exact for all three requests.

Existing deployment registries and audit guidance improve the documentation and review of DP systems [14, 2], while privacy accountants and program-level verification tools answer important but different questions. The restricted object studied in this chapter begins after a run has completed or has been externally reported. Given a frozen evidence bundle $\mathcal{E}$ and a requested wording query q_{clm} , it asks whether the raw-adjacency, transformation, executed-mechanism, and accountant premises jointly entail that exact wording within a finite registry $\mathcal{R}_{\text{val}}$. The corresponding decision $V_{\mathcal{R}_{\text{val}}}(\mathcal{E}, q_{\text{clm}})$ either releases precisely licensed wording or withholds a positive sentence and localizes the unsupported premises.

The decision is evidence-bound in two senses. First, it reopens the selected mapping. It then checks the transformation and preprocessing, the contribution and schedule policies, and the source, runtime, mechanism, and accountant relations needed by the requested claim. Second, its assurance is limited to those observable registered obligations. Digests can expose changed or inconsistent artifacts, but they do not establish that a producer executed the reported code or disclosed every private dependency. Thus the label `EVIDENCE_VALIDATED` denotes a scoped, contract-relative result rather than execution attestation.

The contributions of this chapter are summarized as follows.

- It formalizes a query-dependent, post-execution decision for privacy-unit wording. It establishes the scoped consequence of a registered stability bound from raw adjacency to generated-record distance and identifies the necessity of query separation and the boundary imposed by unauthenticated evidence. It also isolates the factor-of-two mapping that arises when one raw replacement requires one generated-record removal and one addition.
- It instantiates this decision as a finite-registry, fail-closed validator that distinguishes direct, converted, grouped, and underspecified paths. The validator emits a positive sentence only when the requested unit, adjacency, mechanism, runtime, accountant, and numerical consequence all agree.
- It evaluates the complete contract on 100 allowed and 100 paired non-allow cases, five fixed information-necessity witnesses, ten preserved development failures, repeated WISDM and UCI HAR executions, and independent Sepsis and Backblaze mapping diagnostics. Separate verifiers reopen the claim-bearing inputs and recompute the reported decisions. These experiments test conformance and decision separation; they are not estimates of defect prevalence or evidence for a new privacy accountant.

The scope is deliberately narrower than end-to-end DP certification. This chapter introduces neither a new sampler nor a new accountant or group-privacy theorem, and it does not verify arbitrary code or attest faithful execution. Its role in the dissertation is diagnostic and decision-oriented: it determines which wording a completed execution can support, which weaker claim can be recovered through a registered consequence, and when the available model must not be reused for the requested statement without a new native-unit analysis or execution.

2.2 Background and Problem Setting

2.2.1 Privacy Units and Windowed DP-SGD

Differential privacy is defined relative to a domain of datasets and a neighboring relation on that domain. Let $\mathcal{M}$ be a randomized mechanism and let $\mathcal{D} \simeq_{\text{raw}} \mathcal{D}'$ denote two raw datasets that are adjacent under the declared relation. For $\varepsilon \geq 0$ and $0 \leq \delta < 1$, the mechanism is (ε, δ) -differentially private if, for every adjacent pair and every measurable output event B ,

$$\Pr[\mathcal{M}(\mathcal{D}) \in B] \leq e^\varepsilon \Pr[\mathcal{M}(\mathcal{D}') \in B] + \delta. \quad (2.1)$$

The neighboring relation determines the privacy unit and the change against which the probability ratio is bounded [1, 4]. For example, add/remove adjacency compares datasets that differ by the presence of one unit, whereas replace-one adjacency compares fixed-size datasets in which one unit is substituted. A unit may be a generated example, a raw event, a complete owner trajectory, or another explicitly declared contribution. These choices are not interchangeable metadata: changing either the unit or the adjacency changes the mathematical statement in Equation (2.1).

DP-SGD is a mechanism family designed to instantiate this definition on a specified training-record domain. It samples records, clips each selected contribution, adds Gaussian noise to the aggregate, and composes the privacy loss over optimization steps [13, 3]. Its accountant is meaningful only for the sampling rule, sensitivity convention, adjacency, and execution schedule supplied to that accountant. Thus a record-level accountant does not by itself imply event-, patient-, or owner-level privacy when those units have first been expanded into multiple training records.

This distinction is central to sequential learning. Let the raw dataset $\mathcal{D}$ contain timestamped events and let a transformation $\mathcal{T}$ construct the generated training-record

collection $\mathcal{G}$. A generated record $r \in \mathcal{G}$ may be a sliding window whose values depend on the raw-event set $\text{supp}_{\text{evt}}(r)$, and $a_{\text{own}}(r)$ denotes the set of owners attributed to that record. For a frozen mapping export, the largest observed event and owner incidences are

$$\begin{aligned}\mathcal{T}(\mathcal{D}) &= \mathcal{G}, \\ \kappa_{\text{event}} &= \max_e |\{r \in \mathcal{G} : e \in \text{supp}_{\text{evt}}(r)\}|, \\ \kappa_{\text{owner}} &= \max_u |\{r \in \mathcal{G} : u \in a_{\text{own}}(r)\}|.\end{aligned}\tag{2.2}$$

These quantities describe the exported incidence structure; they are not privacy parameters and are not automatically worst-case neighboring-dataset bounds. Even when windows do not overlap, replacing one raw event can replace one generated record. Under a generated add/remove metric, that replacement requires one removal and one addition. Conversely, an insertion may shift window boundaries, change selected identities, or alter a data-dependent schedule, affecting more records than the observed support count suggests. A valid conversion therefore requires a stability statement connecting the declared raw adjacency to the metric used by the executed mechanism and accountant.

Let d_{raw} be the declared raw-dataset metric and let d_{gen} be the generated-record metric at which the base mechanism is accounted. The required registered bound is

$$K \geq \sup_{d_{\text{raw}}(\mathcal{D}, \mathcal{D}') \leq 1} d_{\text{gen}}(\mathcal{T}(\mathcal{D}), \mathcal{T}(\mathcal{D}')).\tag{2.3}$$

The supremum ranges over every raw-adjacent pair in the registered domain, not only the observed dataset. It must cover changes to record identities, inclusion, values, preprocessing, selection, contribution limits, and schedules whenever those components can depend on the protected input. Only after this bound has been established can a generated-record guarantee be interpreted for the requested raw unit.

⌘ 2.1: Raw adjacency and candidate generated-record distance bounds

Declared raw change	Effect on an incident generated record	Base accountant metric	Candidate distance or action
Fixed-slot replacement	Generated-record replacement	replace-one	$K = \kappa$ after complete-support validation
Fixed-slot replacement	Generated-record replacement	add/remove	$K = 2\kappa$: one removal and one addition
Insertion or deletion with fixed support	Generated-record addition or removal	add/remove	$K = \kappa$ after fixed-support validation
Insertion or deletion that shifts, reselects, or adaptively creates windows	Potentially many identity and value changes	either	Establish a bound over all branches; otherwise block

Table 2.1 separates four cases that can appear similar in an exported mapping but support different consequences. In particular, the factor of two follows from the pair of adjacency metrics, not from window overlap alone. The next subsection uses this distinction to formulate evidence-to-claim misalignment: a claim may name a raw unit that is not licensed by the transformation and accounting evidence attached to the completed run.

2.2.2 Evidence-to-Claim Misalignment

Let $\mathcal{E}$ denote a frozen evidence bundle for one completed or externally reported execution, and let q_{clm} denote one requested public sentence. The query specifies the privacy unit and adjacency to be named, together with the numerical pair to be released. Evidence-to-claim misalignment occurs when this interpretation is not jointly entailed by the evidence, even if an accountant value recorded inside $\mathcal{E}$ is internally valid. For example, a pair computed for generated-record add/remove adjacency can be correct at its base domain while remaining insufficient for a sentence that names one raw event or one complete owner. The problem is therefore semantic as well as numerical: the wording determines which neighboring change the public guarantee purports to cover.

Validating that wording requires a connected chain of premises,

$$\begin{aligned} \text{raw adjacency} &\longrightarrow \text{stable transformation} \longrightarrow \text{executed mechanism} \longrightarrow \text{accountant} \\ &\longrightarrow \text{requested wording.} \end{aligned}$$

The first relation fixes how a raw-adjacent change propagates through window construction, selection, and contribution control. The second identifies the sampler, clipping rule, noise mechanism, and schedule that actually consumed the generated records. The third relates those runtime quantities to the accountant implementation and its parameters. The final relation checks that the unit, adjacency, and numbers in

q_{clm} are precisely the consequence licensed by the preceding relations. A downstream match cannot repair an upstream gap: a correctly recomputed accountant value does not establish a raw-unit claim when the transformation bound in Equation (2.3) is absent or uses a different adjacency.

Accordingly, the evidence cannot be limited to a model file and a privacy pair. It must expose the generated-record identities and their raw support or owner attribution, the preprocessing and selection policies, contribution caps and schedules, runtime sampler and optimization parameters, and the mechanism and accountant configurations against which the claim is checked. Components often described as “fixed” require particular care. Labels, normalization statistics, initial model state, record selection, or a later schedule may depend on protected data or on earlier private outcomes. Each such dependency must be public or local to the protected unit, included in the privacy analysis, or covered by a registered worst-case bound. Otherwise the corresponding edge in the chain remains unsupported. Artifact digests can reveal substitution, inconsistency, or staleness among disclosed files, but a matching digest alone cannot prove that the producer executed those files or disclosed every private dependency.

The explicit query is necessary because the same $\mathcal{E}$ can yield different decisions. A window-level request may directly reuse the base pair; an event-level request may be allowed only after applying a registered distance K ; and an owner-level request may have to be blocked when the resulting bound is vacuous or when no complete owner-contribution bound is registered. Thus q_{clm} is not a formatting option appended after validation. It is a scientific input that selects the proposition to be evaluated. A single query-oblivious label attached to $\mathcal{E}$ would erase this distinction and could authorize mutually non-equivalent sentences.

This object is related to, but narrower than, several existing assurance activities. Deployment registries, implementation guidance, and structured privacy challenges

improve artifact disclosure and expert review [15, 2, 14]. Program-level verification and cryptographic proof systems instead ask whether a prescribed computation or training procedure satisfies, or verifiably follows, a DP construction [16, 17]. Mechanisms designed directly for user-level adjacency can also reduce unit mismatch at design time [18]. These lines provide stronger answers to their respective questions, but none makes the requested post-execution wording implicit.

The validation object used here is the joint pair $(\mathcal{E}, q_{\text{clm}})$. Within a finite registry $\mathcal{R}_{\text{val}}$, the validator either emits the exact licensed wording or blocks the positive claim and reports the failed relations. It does not rank arbitrary evidence quality, prove arbitrary programs, recompute a new privacy analysis, or attest execution. Section 2.3 formalizes this query-dependent decision and the fail-closed boundary that keeps unsupported interpretations from being released.

2.3 Post-Execution Evidence-Bound Validation

2.3.1 Typed Evidence Contract and Query-Dependent Decision

A post-execution validator must preserve more information than a Boolean statement that an evidence bundle “passes.” A positive result may follow directly at the accountant’s native unit, may require a registered conversion to another unit, or may be unavailable because the corresponding derivation is absent or vacuous. Collapsing these cases would hide both the semantics of the released privacy pair and the reason that a nearby request was withheld. The primary decision object is therefore a typed tuple. For finite registry $\mathcal{R}_{\text{val}}$, frozen evidence $\mathcal{E}$, and requested claim q_{clm} , define

$$V_{\mathcal{R}_{\text{val}}}(\mathcal{E}, q_{\text{clm}}) = \left(\text{path}, \text{release}, \text{assurance}, K, \widehat{\varepsilon}, \widehat{\delta}, \mathcal{I} \right). \quad (2.4)$$

The path field identifies the licensed relation between the base accounting unit and

the unit requested in q_{clm} . **Direct** denotes aligned units and metrics, so the registered base pair is retained. **Convert** denotes the registered event-conversion path, whereas **Group** records an owner- or group-level derivation. **Underspecified** is used when the requested interpretation has no complete registered path. These labels describe how a claim would be derived; they do not themselves authorize a public sentence. In particular, a **Group** path can still be blocked when its numerical consequence is vacuous.

The release field is evaluated separately from the path. Its exact states are **Allowed**, **Blocked-Invalid**, **Blocked-Unverified**, **Blocked-Vacuous**, and **Research-Only**. An allowed result requires a registered path, agreement among the disclosed execution and accountant fields, finite nonvacuous numbers, and exact unit-and-adjacency wording. Invalid values or contradictions produce **Blocked-Invalid**; a missing observable obligation produces **Blocked-Unverified**; and a numerically empty group consequence produces **Blocked-Vacuous**. A research-randomness or external-intake record can be retained as **Research-Only** without becoming public DP wording. Every blocked public view may report issue codes and forbidden wording, but it contains no copy-ready positive privacy sentence.

The assurance field records the scope and authority of the decision, not a confidence score. It is exactly **EVIDENCE_VALIDATED**, **EXTERNAL_UNVERIFIED**, or **DIAGNOSTIC_ONLY**. In this chapter, **EVIDENCE_VALIDATED** means that the observable obligations registered for the selected path have passed; it does not upgrade file consistency to execution attestation. The fields $K, \widehat{\varepsilon}, \widehat{\delta}$ retain the registered distance and the contract-permitted numerical consequence. Their presence in diagnostics does not override a blocked release, while $\mathcal{I}$ localizes every blocking or qualification condition. Keeping path, release, and assurance separate prevents an allowed conversion, a blocked group derivation, and an unattested execution from being represented by the

same undifferentiated status.

Table 2.2 states the six claim-bearing surfaces checked together. No single row repairs a contradiction in another: an accountant cannot authenticate the mapping, a mapping digest cannot establish runtime sampling, and a correctly formatted public sentence cannot supply a missing stability proof.

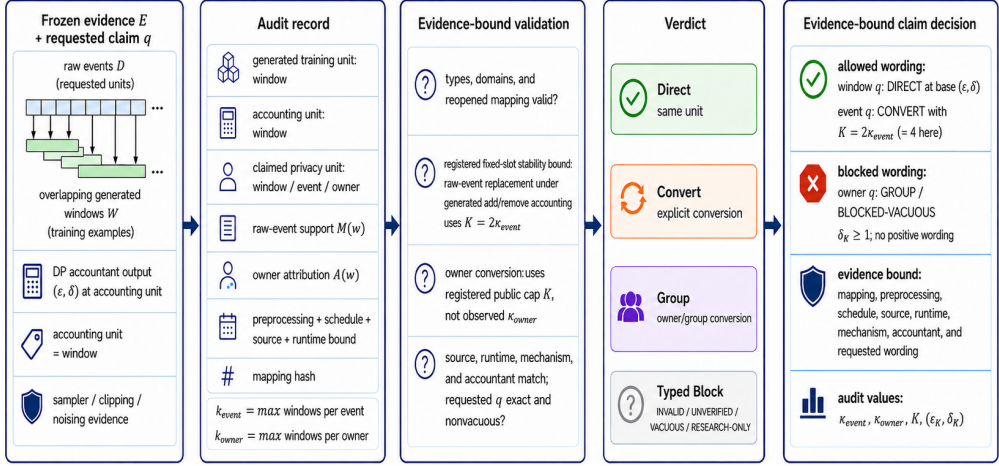


그림 2.1: Evidence-bound validation and query separation for one frozen WISDM execution

Lemma 2.1 (Query separation). *For fixed $\mathcal{E}$, suppose two requested claims satisfy $V_{\mathcal{R}_{\text{val}}}(\mathcal{E}, q_1) \neq V_{\mathcal{R}_{\text{val}}}(\mathcal{E}, q_2)$. No query-oblivious interface $A(\mathcal{E})$ can be exact for both requests.*

Proof. The interface receives the same input for both requests and therefore has one output distribution. That distribution cannot be concentrated on two distinct decision tuples. Returning one common result is therefore incorrect for at least one request; a common abstention is likewise inexact whenever either request is licensed. Hence exactness requires the requested claim to be supplied as an input. $\square$

표 2.2: Claim-bearing evidence contract for one post-execution decision

Surface	Required evidence	Decision role
Query contract	Requested raw unit and adjacency, assurance level, conversion identifier, and candidate wording	Fixes the exact proposition; there is no context-free epsilon
Actual mapping	Stable generated identity and unit, half-open raw support, complete owner attribution, selected identities, and full/canonical digests	Recomputes counts and incidence instead of trusting a copied summary
Pipeline manifest	Dataset and split, identity and preprocessing policies, contribution and population rules, schedule, selection, and source digests	Exposes dependencies that can enlarge raw-to-generated influence
Mechanism declaration	Sampling, clipping, noise, mechanism and accountant identifiers, versions, unit, adjacency, sample rate, and steps	Defines the base DP statement that a registered path may transport
Runtime trace	Observed mechanism fields, mapping and pipeline bindings, update trace, released-model digest, and randomness mode	Checks equality between the declaration and the concrete reported run
Public report	Path, release and assurance states, validated K , base/reported pair, assumptions, issues, supported and forbidden wording	Separates copy-ready claims from diagnostics and explicit abstention

Lemma 2.1 is an interface-separation result, not a new privacy theorem. It establishes why the query in Equation (2.4) cannot be replaced by a display option applied after an evidence-only verdict. Figure 2.1 illustrates the consequence: one frozen execution can license two different positive pairs and block a third sentence. The next subsection specifies how a registered distance K determines the converted pair and when the vacuity gate forces **Blocked**.

2.3.2 Privacy-Unit Conversion and Claim Blocking

Once the requested path has been identified, a conversion is licensed only by a registered distance bound connecting the two adjacency relations. Suppose the bound execution is established as (ε, δ) -DP at generated-record distance one and Equation (2.3) holds for an integer $K \geq 1$. Standard group privacy then gives

$$\varepsilon_K = K\varepsilon, \quad \delta_K = \delta \sum_{i=0}^{K-1} e^{i\varepsilon}. \quad (2.5)$$

Equation (2.5) is a registered consequence of the base guarantee, not a new accountant or privacy theorem [1]. The validator neither estimates K from the requested wording nor substitutes an observed incidence for a worst-case bound. It first requires a registered transformation argument, then evaluates the geometric sum with stable log-domain arithmetic. For $\varepsilon > 0$, the implementation uses

$$\log \delta_K = \log \delta + \log(\text{expm1}(K\varepsilon)) - \log(\text{expm1}(\varepsilon)), \quad (2.6)$$

with the exact $\varepsilon = 0$ limit $\delta_K = K\delta$. This avoids overflow without changing the privacy consequence. A public pair is emitted only when both values are finite, $\varepsilon_K \geq 0$, and $0 < \delta_K < 1$. A non-finite result, $\delta_K \geq 1$, or an unregistered derivation blocks positive wording. The resulting values may remain visible as diagnostics, but their

presence does not convert a vacuous result into an allowed privacy claim; clamping delta to one for serialization is not a privacy result.

Proposition 2.1 (Contract-relative consequence). *Assume that (i) $\mathcal{E}$ describes $\mathcal{T}$ and the executed mechanism $\mathcal{M}$; (ii) a registered checker establishes*

$$d_{\text{gen}}(\mathcal{T}(\mathcal{D}), \mathcal{T}(\mathcal{D}')) \leq K$$

*for every pair satisfying $d_{\text{raw}}(\mathcal{D}, \mathcal{D}') \leq 1$; and (iii) a registered accountant establishes (ε, δ) -DP for the bound execution at generated-record distance one. Then $\mathcal{M} \circ \mathcal{T}$ satisfies Equation (2.5) for the requested raw adjacency. If $K = 1$ and the units and metrics coincide, the base pair takes the *Direct* path.*

Proof. For a raw-adjacent pair, premise (ii) supplies a generated-metric path of length at most K between the two transformed datasets. Applying the distance-one DP inequality successively along that path multiplies the likelihood factor by e^ε at each step. The additive terms accumulate with multipliers $1, e^\varepsilon, \dots, e^{(K-1)\varepsilon}$, which yields Equation (2.5). When $K = 1$, the expression reduces to the base pair. The conclusion is conditional on the three registered premises; the validator checks their disclosed evidence rather than proving execution fidelity. $\square$

The metric pair is essential when selecting K . A raw replacement can replace several generated windows, but a replacement is one edit under a generated replace-one metric and two edits under a generated add/remove metric. This distinction produces the factor-of-two case summarized in Table 2.1.

Corollary 2.1 (Factor-two mapping). *Assume public fixed-slot identities and a registered complete-support bound under which one raw-event replacement changes at most κ_{event} generated records. A generated add/remove accountant licenses $K = 2\kappa_{\text{event}}$, whereas a generated replace-one accountant licenses $K = \kappa_{\text{event}}$.*

Proof. Each affected generated record is replaced. Under add/remove distance, one such replacement consists of one removal and one addition, so at most $2\kappa_{\text{event}}$ edits are required. Under replace-one distance, each replacement costs one edit, giving κ_{event} . The conclusion relies on the registered complete-support premise; observed overlap alone is insufficient. $\square$

These consequences produce four distinct decision cases. A native-unit request with coincident adjacency uses $K = 1$ and retains the base pair as **Direct**. A raw-event request takes **Convert** only when the event stability argument and its metric-specific K are registered. An owner request takes **Group** when a public contribution bound supplies the generated distance. If no such bound is registered, the path is **Under-specified**; if the bound exists but Equation (2.5) is non-finite or vacuous, the path remains typed but the release is **Blocked**.

The WISDM evidence in Figure 2.1 illustrates why a public cap and an observed maximum are not interchangeable. Its fixed-slot event replacement has $\kappa_{\text{event}} = 2$ and generated add/remove accounting, so Corollary 2.1 gives $K = 4$. For the owner path, a public prefix policy caps contributions at 600 windows before training. The observed maximum of 565 verifies compliance with that policy but does not replace it as a neighboring-dataset bound; the conversion therefore uses $K = 600$. Because the resulting δ_K is at least one, the owner path is reported as **Group/Blocked** and no positive owner-level sentence is released.

The conversion rule thus determines what follows if the evidence premises hold; it does not establish that the disclosed mapping, runtime, mechanism, and accountant all belong to the same execution. The next subsection specifies those binding obligations and the assurance boundary of a bundle-only decision.

2.3.3 Decision Procedure and Artifact Invalidation

The conversion consequence in Section 2.3.2 is meaningful only when the transformation, runtime, mechanism, and accountant records refer to the same reported execution. The validator therefore binds more than a model artifact and a final privacy pair. Its internal record includes the selected mapping and identities; event support and owner attribution; preprocessing, population, contribution, selection, and schedule policies; mechanism, runtime, source, and accountant versions; base privacy parameters; and the requested-unit contract. The public view may redact private provenance, but it retains the decision, artifact bindings, numerical result or diagnostic, assumptions, forbidden wording, localized issues, and conditions that trigger revalidation.

The finite-registry validator applies the following eight phases in order. This procedure is a typed evidence-to-wording decision, not a training algorithm.

- (1) **Types and domains.** Exact typed parsing rejects Boolean values supplied as integers, non-finite numerical values, invalid probabilities, and fields outside the registered domain before any privacy consequence is evaluated.
- (2) **Reopened mapping.** The validator reopens the mapping rather than trusting a copied report string. It recomputes file and semantic bindings, selected identities, intervals, event supports, owner attributions, κ_{event} , and κ_{owner} .
- (3) **Registered stability.** The selected checker must cover record identity and influence together with preprocessing, population, contribution, schedule, and selection dependencies. A checker that covers only observed overlap does not establish Equation (2.3).
- (4) **Runtime agreement.** Observed fields must match the registered sampler, sampling rate, optimization steps, noise and clipping rules, accounting unit, mapping

and pipeline identities, and secure-randomness mode. Missing runtime observation blocks concrete-run wording.

- (5) **Registry binding.** A versioned registry binds supported source and adapter identities to the mechanism semantics and accountant. The validator recomputes the base pair and default-denies unknown adapters, attachments, or accounting rules.
- (6) **Path resolution.** The requested unit and adjacency select the direct, converted, grouped, or underspecified path and its registered stability method. The validator never infers a path from the wording alone.
- (7) **Numerical consequence.** The base pair is retained only on the exact native path. Otherwise Equations (2.5) and (2.6) are evaluated under the registered K .
- (8) **Typed release.** Exact wording, issue precedence, the non-finite and vacuity gates, and the authority label determine the public report. The operational metadata may be redacted without changing that decision.

Four complementary digests support these checks. A full-file digest detects byte substitution. A canonical selected digest binds the normalized semantic records that participate in the run, so permutation or irrelevant decoy rows cannot silently redefine the selected mapping. A pipeline digest binds transformation and policy parameters, and a runtime digest binds the observed mechanism fields. These digests expose changed, stale, or mutually inconsistent artifacts. They do not establish that the producer executed the referenced bytes: equal mapping digests do not prove sampler use, and equal report text cannot replace reopening the underlying files. Changing a bound mapping, selected identity, preprocessing or population rule, contribution policy, schedule, mechanism or accountant version, runtime parameter, claim unit, adjacency, or conversion method therefore invalidates the old report and triggers validation

again. A redacted public view is a view of the same decision, not a separately certified claim.

When several conditions apply, the implemented release precedence is

$$\begin{aligned} \text{Blocked-Invalid} &> \text{Blocked-Unverified} > \text{Blocked-Vacuous} \\ &> \text{Research-Only} > \text{Allowed}. \end{aligned} \tag{2.7}$$

The ordering prevents a valid scalar calculation from masking a malformed input or missing runtime trace. Likewise, an external attachment cannot repair an unrelated mechanism contradiction, and a blocked internal diagnostic cannot be converted into an allowed public sentence by redaction.

2.3.4 Evidence Authority, Complexity, and Non-Attestation Boundary

The validation cost is dominated by reopening and checking these structures. For B mapping bytes, R rows, S selected rows, A attribution entries, and $P \leq 2A$ interval endpoints, the registered checks require $O(B + R + A + S \log S + P \log P)$ time and $O(R + A)$ memory, in addition to the work of the registered accountant. This bound describes the complete validator; a specialized aggregate rescan that checks only mapping statistics is not an equivalent certificate.

The remaining assurance boundary is informational. A bundle-only validator can test consistency among the objects it receives, but unauthenticated self-reporting cannot distinguish faithful execution from a producer that presents the same bundle after executing different code or hiding a dependency.

Proposition 2.2 (Bundle-only non-attestation). *Let a validator observe only an unauthenticated tuple $(\mathcal{E}, q_{\text{clm}}, \mathcal{R}_{\text{val}})$. Suppose a faithful world and a world that reports the same tuple while executing different code or hiding a dependency are observationally*

identical to the validator. Every deterministic or randomized validator has the same output distribution in both worlds and therefore cannot establish execution fidelity from that tuple alone.

Proof. The validator receives identical inputs in the two worlds. A deterministic validator must consequently return the same output, and a randomized validator must induce the same conditional output distribution. No decision rule over the observed tuple can separate the worlds without an additional authenticated observation. $\square$

The threat model is consequently an honest-but-fallible producer: stale exports, omitted dependencies, inconsistent configuration, mismatched artifacts, and unjustified privacy-unit restatement are in scope. A malicious producer that can fabricate a mutually consistent bundle while running another procedure is outside the positive assurance claim. Addressing that adversary requires authenticated runtime observation, execution attestation, or a cryptographic proof system [16, 17]. The experiments in the next section test whether the registered observable obligations separate designed allowed and non-allow cases; they do not measure faithful execution. In this chapter, `EVIDENCE_VALIDATED` means that registered observable obligations pass; it does not mean execution fidelity.

2.4 Experimental Evaluation

2.4.1 Controlled Validation and Ablation

The first experimental question is which claim distinctions disappear when the validator observes only a weak projection of the evidence or when a registered obligation is removed. The controlled suite contains 100 allowed cases spanning five registered path strata—direct window, direct event, direct owner, converted event, and grouped

owner—at five boundary or scale levels and in four mapping contexts. The contexts include a canonical mapping, irregular supports, non-selected decoy rows, and permuted file order. Each allowed parent is paired with one non-allow case. The 100 paired cases are produced by 25 one-premise mutation operators, each applied to four distinct parents, so that every allowed parent is consumed once.

An authored manifest, rather than the output of the full validator, declares the expected path, release and assurance state, localized issue, and numerical rule. Every method receives the same report, audit row, and mapping. The *unsafe-positive fraction* is the fraction of the 100 non-allow cases that nevertheless receive an allowed decision. The *valid-exact fraction* is the fraction of the 100 allowed cases for which both the decision path and privacy pair are exact. The *exact-tuple fraction* is evaluated over all 200 cases and additionally requires the correct assurance state, block subtype, and presence or absence of numerical output. These fractions are deterministic coverage outcomes for the authored suite. They are neither estimates of a deployment population nor empirical error rates.

Table 2.3 compares the complete registered contract (FULL) with two kinds of deliberately weakened checks. B0–B4 are information projections that retain only field presence, JSON Schema validity, multiplicity, mapping-hash binding, or base-accountant recomputation. A1 leaves the static contract intact but assumes that runtime evidence matches it, whereas A2 removes only the vacuity gate. The rows are therefore diagnostic projections and obligation ablations, not predecessor systems or competing validators.

The complete contract produces no unsafe positive among the 100 paired non-allow cases, exactly matches all 100 allowed cases, and returns the expected full tuple in all 200 cases. An independent standard-library implementation of the grid and mutation oracle passes 20,909 of 20,909 checks, and the expanded contract suite passes

⌘ 2.3: Controlled-suite and ablation results

Probe / full contract	Unsafe-positive frac.	Valid-exact frac.	Exact-tuple frac.	Principal evidence not checked
B0 Field presence	1.00	0.60	0.00	types, semantics, files, execution, accounting
B1 JSON Schema	0.96	0.60	0.00	cross-field and execution semantics
B2 Multiplicity only	1.00	0.80	0.00	stability/adjacency, staleness, mechanism
B3 Hash only	0.96	0.60	0.00	consistently wrong semantics or mechanism
B4 Accountant only	0.88	0.60	0.00	mapping, claimed unit, dependencies, runtime
A1 Runtime assumed matched	0.12	1.00	0.94	independent runtime/artifact equality
A2 FULL minus vacuity	0.04	1.00	0.98	reportability of converted numerics
FULL	0.00	1.00	1.00	registered-contract assurance boundary

83 of 83 checks. These checks establish conformance to the finite authored contract, not correctness for an unregistered mechanism or unobserved execution.

The weaker rows expose distinct losses of information. B0, B1, B3, and B4 reuse the base privacy pair in 40 allowed conversion or grouping cases, while B2 does so in four. A1 retains every allowed case but leaves 12 unsafe positives: four each for missing-trace, runtime-noise, and runtime-artifact mutations. A2 leaves four vacuous owner-level positives. Five fixed necessity witnesses isolate the corresponding obligations: reopened mapping bytes, registered stability status, runtime noise and trace equality, the vacuity gate, and exact event rather than window wording. Removing each obligation erases the decision boundary between its paired valid and blocked witness. Independent recomputation passes all 10 witness decisions as well as their associated tamper tests.

Table 2.4 is intentionally narrower than a universal minimality claim. The witnesses show that deleting one named edge loses a required distinction on authenticated fixed inputs; they do not prove that every audit engine must use the same files or that such failures occur at a measured rate in published systems.

A second authority source avoids relying only on newly authored mutations. A protocol fixed before the conformance construction identified ten preserved development incidents whose root causes affected claim-bearing output and had pre-protocol discovery records. A production-import-free verifier reopens the historical source or archive member, discovery anchor, consequence, and 16 repaired controls, passing 176 of 176 checks. Three tamper tests retain the authentic lineage and reject an omitted incident or forged archive identity. This is actual development lineage from the evaluated system, but it is not an independently sampled publication corpus and therefore does not estimate real-world prevalence.

For scope comparison, the official TensorFlow Privacy 0.9.0 privacy-statement

표 2.4: Fixed information-necessity witnesses

ID	Fixed comparison	Deleted or hidden edge	Required distinction
W1	Valid parent P01 vs. blocked F02	Actual mapping bytes; report, audit row, and query held equal	Allowed vs. Blocked-Invalid
W2	Converted parent P02 vs. F16	Registered stability status; remaining public fields held equal	Convert/Allowed vs. Blocked-Unverified
W3	Direct parent P01 vs. F05	Runtime noise and trace equality	Direct/Allowed vs. Blocked-Invalid
W4	F01 under A2 vs. FULL	Vacuity and reportability gate	Unsafe owner positive vs. typed vacuous block
W5	One WISDM event certificate	Exact licensed event sentence replaced by a valid window sentence	Independent wording verifier rejects the substitution

function [19] is evaluated on the original 23-case subset that can be expressed through its scalar interface. Given caller-supplied scalars and a group bound, it returns finite conditional statements for all seven valid cases and for 15 of 16 non-allow bundles; it rejects a Boolean value supplied for the number of steps. All 79 reproduction checks re-execute and hash the official module and setup. In the fixed-base vacuity case, the function solves a smaller example-level delta and freshly targets a user-level delta. This is a useful alternative accounting contract, but it is not Equation (2.5) applied to the supplied privacy pair. Accordingly, the comparison is not a quality ranking: the scalar function derives a conditional statement from values supplied by its caller, whereas FULL reopens the pipeline evidence and decides whether the exact submitted tuple supports the requested sentence. The next subsection tests that evidence-bound distinction on activity data and mapping-only rescans.

2.4.2 Complete Activity-Data Executions

The second experimental question asks whether one frozen evidence bundle produces distinct query-dependent decisions and whether that pattern recurs across fresh, separately bound executions. The complete activity-data path uses the public WISDM accelerometer data [20]. A public parser reduces 1,098,210 physical lines to 1,086,465 canonical valid events. From these events, the registered transformation constructs 7,304 training windows, each with length 200 and stride 100. Owner identities partition the mapping, feature and label transforms are record-local, and a public prefix policy caps training-window contributions at 600 per owner before training. The bound run uses Poisson window sampling at $1/50$, 100 optimization steps, and noise multiplier 2. Registered RDP recomputation yields the base pair $(0.536827391167, 10^{-6})$.

Every request in Table 2.5 reopens the same 7,304-row mapping and binds the same pipeline and runtime evidence. The window query is native to the accountant and

⌘ 2.5: Query-dependent decisions for one frozen WISDM execution

Requested claim	Observed / registered (K)	Path / release	Pair or action
Window	– / 1	Direct / Allowed	$(0.53683, 10^{-6})$
Raw event	2 / 4	Convert / Allowed	$(2.14731, 1.0642 \times 10^{-5})$
Owner	565 / 600	Group / Blocked	vacuous: $\delta_K \geq 1$

therefore uses $K = 1$. Under fixed-slot raw-event replacement, the reopened mapping gives $\kappa_{\text{event}} = 2$; because the base metric is generated add/remove, Corollary 2.1 gives $K = 4$. The owner request observes a maximum contribution of 565 only as a compliance check. Its registered conversion uses the public cap $K = 600$, not that observed maximum.

Thus the same execution licenses the base window sentence, licenses a weaker event-level sentence after registered conversion, and withholds positive owner-level wording. The owner path remains **Group**; its release is separately **Blocked** because Equation (2.5) produces $\varepsilon_K = 322.096$. Its converted delta satisfies $\log_{10} \delta_K = 134.033097$, and hence $\delta_K \geq 1$. The existence of those diagnostic numbers does not make them reportable privacy parameters.

Five post-freeze WISDM runs have distinct model digests and preserve all three decision paths and privacy values. Across those runs, accuracy is 0.6556 ± 0.0080 and macro-F1 is 0.3922 ± 0.0033 . A separately bound The complete pipeline—raw parsing, mapping, training, report generation, and evidence production—takes 16.743 ± 1.191 seconds over the five runs in the reported environment. Independent verification with raw reconstruction takes 6.480 ± 0.428 seconds. Five additional portable invocations take 1.577 ± 0.125 seconds and pass 35 of 35 checks each while omitting the three

raw-input reconstruction checks. A complete run directory averages 0.996 MB; the separately acquired 50.326 MB raw file is required only for full-source reconstruction. These end-to-end timings include process startup and are not isolated validator overhead.

A separately bound route uses the UCI Human Activity Recognition Using Smartphones data [21, 22], with 7,352 training windows, 2,947 test windows, and 561 features. Each of five post-freeze runs passes 49 of 49 raw-source checks and 38 of 38 portable checks with a distinct digest. Because event support and an owner contract are absent, each run licenses only **Direct** window add/remove wording at approximately $(0.536827, 10^{-6})$; event and owner requests are blocked rather than inferred. Its mean accuracy is 0.8049 ± 0.0315 , and mean macro-F1 is 0.7795 ± 0.0504 . Both datasets' accuracy and F1 values are integration diagnostics, not competitive or state-of-the-art utility claims.

⌘ 2.6: Invariant UCI HAR authorization boundary across five executions

Requested unit	Release status	Path	<i>K</i>
Published window	Allowed	Direct	1
Raw event	Blocked- Unverified	Underspecified	–
Owner	Blocked- Unverified	Underspecified	–

Table 2.6 is invariant across all five protocol-fixed runs. This negative result is informative: an existing published-window dataset does not automatically expose the raw-session support or public owner cap needed for a stronger unit claim, even when the DP-SGD mechanism and accountant are otherwise reproduced.

2.4.3 Sepsis and Backblaze Mapping Diagnostics

The third experimental question isolates the transformation side of the contract at substantially larger scales. For the PhysioNet/Computing in Cardiology 2019 Sepsis data [23, 24, 25], the verifier opens and hashes 5,000 patient files and reconstructs 16 mappings under dense, non-overlap, and public-cap policies. For the Backblaze first-quarter 2025 drive statistics [26], a verifier without production imports rescans 1.02 GB comprising 90 daily files, 27,799,986 drive-day rows, and 318,426 drives. It distinguishes windows over ordered available snapshots, which may compress calendar gaps, from calendar-contiguous windows.

The Sepsis source contains 192,206 hourly patient rows. Six-, twelve-, and twenty-four-hour window policies cover dense, half-stride, and non-overlap schedules, while public owner-local prefix caps of 1, 4, 8, and 16 make the contribution rule observable before selection. Label-balanced caps act as negative controls: they can reproduce the same displayed counts as a public prefix cap while using private labels to change selected identities. Candidate event distance is therefore reported only for fixed positional or public-prefix policies whose support premise is registered.

The Backblaze rescan covers 27,799,986 unique drive-day events with no duplicate serial-number and date pair and evaluates 22 mapping policies. Ordered-snapshot and calendar-contiguous windows can display equal incidence while selecting different records; public owner-prefix caps are evaluated separately. This distinction is why a compact table can summarize representative rows but cannot replace the full policy identity bound into the mapping evidence.

The Sepsis and Backblaze verifiers pass 128 of 128 and 86 of 86 checks, respectively. Table 2.7 nevertheless exposes three distinct premise risks. First, non-overlap can reduce κ_{event} to one while fixed-slot replacement under generated add/remove accounting still gives candidate $K = 2$. Second, ordered and calendar-contiguous

⌘ 2.7: Mapping-only diagnostics for Sepsis and Backblaze data

Dataset	Mapping policy	Generated records	Obs. κ_{event}	Candidate event K	Obs. κ_{owner}
Sepsis	6h dense	167,206	6	12	331
Sepsis	6h non-overlap	29,958	1	2	56
Sepsis	public owner cap 1	5,000	1	2	1
Backblaze	7d dense, ordered snapshots	25,904,370	7	14	84
Backblaze	7d dense, calendar-contiguous	25,867,949	7	14	84
Backblaze	7d non-overlap, ordered	3,709,484	1	2	12
Backblaze	7d non-overlap, calendar	3,707,669	1	2	12
Backblaze	60d non-overlap, ordered	304,906	1	2	1
Backblaze	60d non-overlap, calendar	302,760	1	2	1

Backblaze semantics differ by 36,421 dense seven-day windows despite equal incidence. Third, a Sepsis label-balanced selection can reproduce public-cap counts while privately changing selected identities. The rescans reconstruct rows and policies, but neither bundle binds a mechanism, runtime, or accountant. Consequently, every row is only a mapping diagnostic: no row is an (ε, δ) certificate or a **Direct** verdict, and no positive privacy wording is emitted.

2.4.4 Independent Verification, External Intake, and Reproducibility

The final experimental question asks whether each result can be reconstructed without merely importing the production validator. Separate implementations reopen the relevant inputs, recompute hashes, rows, decisions, and numerical tolerances, and confirm that a blocked public output contains no positive sentence. Table 2.8 reports the principal surfaces. The denominators enumerate deterministic assertions in fixed

표 2.8: Independent verification surfaces and stated authority

Surface	Passing checks	Authority
Expanded contract regression	83/83	Fixed schema, registry, and release semantics
Designed 200-case conformance surface	20,909/20,909	Five strata, 25 operators, fixed oracle and lineage
Necessity witnesses	10/10	Five fixed edge-deletion distinctions
Retrospective development lineage	176/176	Ten preserved incidents and 16 repaired controls
TensorFlow Privacy scope comparison	79/79	Re-execution and hashing of the official scalar interface
WISDM reference and each confirmatory run	38/38	Full raw reconstruction and three query decisions
UCI HAR full / portable, each run	49/49 / 38/38	Published-input reconstruction / distributable evidence
Sepsis / Backblaze mapping rescans	128/128 / 86/86	Mapping and policy diagnostics only

artifacts; they are not confidence intervals or independent human replications.

An externally authored ExtraSensory source-path intake provides a separate authority test [27, 28]. Six terminal records are retained rather than reporting only the successful attempt: repository import, path, disk, loader, and plotting failures precede one complete five-epoch execution. The final run produces all 17 expected outputs, and separate verifiers reconstruct its mapping and execution records. Nevertheless, the external source sets `secure_rng=False` and prints a configured target epsilon rather than an accountant-derived achieved epsilon. The terminal result is therefore execution-complete but claim-unverified; positive DP wording remains empty. This

case demonstrates why successful training and artifact presence are not sufficient authority for a privacy sentence.

The reproducibility package pins the core Python dependencies, schemas, registry, claim-bearing reports, frozen outputs, and verification entry points. Every reported table row has a machine-readable source, while larger public datasets are identified by acquisition boundary and digest rather than redistributed. Exact frozen-number verification does not require retraining; fresh WISDM runs use new private randomness, so the verifier requires internal consistency and invariant decision semantics rather than identical model or batch-trace hashes. No nondeterministic outcome is silently dropped, and missing raw inputs, unregistered adapters, research randomness, private-dependent selection, and vacuous conversion remain typed limitations rather than being converted into positive results.

2.5 Summary

This chapter treated a privacy statement as a query over frozen execution evidence rather than as a synonym for an accountant output. The validator first checks whether the requested unit and adjacency are connected to the generated records, mechanism, runtime, and accountant through a registered evidence chain. It then reports a typed path and independently decides whether positive wording is allowed. As a result, one completed execution can support a native-unit claim, support a weaker converted claim, and block a stronger claim without collapsing those outcomes into one privacy label.

The resulting assurance is deliberately bounded. It is relative to the supplied evidence and a finite validation registry; it is not execution attestation, an arbitrary-code proof, or a new accountant. Reopened artifacts, semantic digests, runtime equality,

registered stability, recomputation, and fail-closed release reduce accidental claim–evidence mismatches under the honest-but-fallible threat model. They cannot distinguish faithful execution from a producer that presents a consistently fabricated bundle. Missing, unregistered, non-finite, or vacuous premises therefore suppress positive wording rather than justify extrapolation.

The practical conclusion is a reuse boundary. A completed model may be reused with the exact native or converted wording licensed by its evidence, but a blocked patient- or owner-level request cannot inherit the window-level pair. Satisfying that stronger request requires additional registered premises or a new mechanism and execution native to the requested unit. Chapter 3 turns from privacy wording to generative-image provenance. The assurance object changes, but the same dissertation-level question remains: whether a public decision is supported by the target, mechanism, interface, and evidence that the decision claims to represent.

제 3 장 Secure Public Verification of Generative-AI Provenance: PP-Mark

3.1 Introduction

The preceding two chapters developed the Auditable Privacy axis of this dissertation. They asked whether a privacy sentence is licensed by a completed execution and whether a requested privacy route is coherently assembled before that execution begins. This chapter turns to a separate trust claim: whether a third party can verify that a generated image is accompanied by evidence consistent with a declared generation context. The protected object, mechanism, and evidence are different, but the assurance problem has the same form. A positive label should not be released merely because one observable statistic crosses a threshold; the label must remain aligned with the process and evidence that give it meaning.

Image provenance has become harder to establish as generative models have improved. Synthetic images can support benign creation, accessibility, and simulation, but they can also be used for impersonation and misleading media. The resulting provenance problem is not solved by visual inspection alone [29, 5]. Content credentials and signed metadata provide an important publication layer, yet the strength of a provenance claim still depends on what the attached technical evidence actually establishes [6]. A signature can authenticate a manifest supplied by a key holder. It does not, by itself, show that a registered watermark signal was embedded according to the process named in that manifest.

Digital watermarking addresses this gap by placing a recoverable signal in the image or in the generative process. Post-hoc methods encode a message after an image is

produced, while latent and model-integrated methods introduce structured signals during generation [30, 31, 32, 33]. Verification commonly evaluates a detector $D(I, \text{md})$ and accepts when its score exceeds a calibrated threshold. This design offers low latency and can remain useful after moderate compression or transformation. Its weakness is also structural: once the detector is public, the same decision surface that enables independent checking becomes a target for queries, surrogate construction, or gradient-based optimization [7, 8, 9].

Keeping the detector private behind an application programming interface reduces direct exposure but replaces technical verification with organizational trust. Only the service operator can execute the hidden rule, rate-limit queries, or resolve disputes. A user cannot reproduce the judgment independently, and a long-lived provenance record remains dependent on the continued availability and integrity of one verifier. Public verification and forgery resistance therefore appear to pull in opposite directions: opening the detector improves auditability but expands the attack surface, whereas closing it reduces independent auditability.

This chapter studies PP-Mark, a construction that changes the acceptance rule rather than merely hiding or hardening the score. The generator derives a latent watermark from a secret-keyed binding to the prompt, seed, model identifier, and resolution. It commits a deterministic sample of the embedding trace under a Merkle root and produces an SP1 zero-knowledge proof receipt. A public verifier recovers a statistical watermark score from the submitted image and verifies the receipt against the declared context, the committed trace, and the exact canonical image hash. Full acceptance requires both conditions:

$$\begin{aligned} \text{PPMark}_{\text{accept}}(I, \text{md}, \pi) &= 1 \\ \iff (S_{\text{det}}(I, \text{md}) \geq \tau) \wedge (\text{VerifyReceipt}(\pi; x_{\text{pub}}) &= 1). \end{aligned} \tag{3.1}$$

Equation (3.1) is the chapter’s central claim-alignment boundary. The score provides image-side evidence that the expected signal is recoverable. The receipt provides process-side evidence that the opened committed trace is consistent with a secret-key-derived binding and an image-bound challenge. Neither term is promoted into the other. A high score alone is a screening result, not publicly verified provenance, and a valid receipt alone does not show that the published pixels retain the detectable signal.

The contributions of this chapter are summarized as follows.

- It formulates secure public provenance verification as a conjunctive evidence decision. The public detector remains available for low-latency screening, while a stronger acceptance mode additionally requires a cryptographic receipt bound to the declared generation context and exact image.
- It defines a context-bound latent watermark. A secret-keyed binding determines a Reed–Solomon payload, sampled latent positions, Gaussian baselines, and signed perturbations. The same binding therefore anchors the statistical sign pattern and the proof predicate.
- It commits sampled trace tuples and verifies a randomized opening in an SP1 zkVM. The construction states the public/private split explicitly and gives a conditional unforgeability bound containing hash, proof-system, and partial-opening failure terms.
- It evaluates the two verification modes against fabricated bindings, compliance bypass, black-box imprint transfer, white-box optimization, regeneration, steganalysis, and common image transformations. It also records the proving, verification, receipt-size, quality, and two-model limits needed to interpret the result operationally.

The assurance boundary is deliberately narrower than proof of image truth or authorship. PP-Mark does not determine whether the prompt is factually correct, whether the depicted event occurred, whether a person owns a legal identity, or whether every diffusion operation was faithfully executed. Its circuit does not re-simulate the full diffusion model. It verifies the declared composite relation over a committed sampled trace, and full acceptance is meaningful only under the stated hash, proof-system, key-secrecy, predicate, statistical, and partial-opening assumptions. The correct conclusion is therefore that an accepted image carries verifiable provenance evidence relative to the declared binding, not that the image is unconditionally authentic.

3.2 Background and Related Work

3.2.1 Generative Image Watermarking and Provenance

An image-watermarking system normally exposes three logical procedures. An embedding procedure receives an image or generation state and a payload; an extraction procedure recovers a message or statistic from a candidate image; and a decision procedure compares the recovered object with a reference or threshold. These procedures can be placed at different points in the image pipeline.

Post-hoc watermarking changes the completed pixel image. HiDDeN learns an encoder–decoder pair for hiding data in images, StegaStamp targets robustness under physical and geometric transformations, and more recent systems such as TrustMark and InvisMark improve capacity or robustness at arbitrary image resolutions [30, 31, 34, 35]. Post-hoc deployment is attractive because it can wrap an existing generator. At the same time, the watermark is separated from the internal generation path and may be attenuated by regeneration or model-based removal [36].

Generative or semantic watermarking moves the signal into the model or latent

trajectory. Tree-Ring introduces structured patterns in initial diffusion noise, Gaussian Shading encodes a lossless latent distribution, and Stable Signature roots a watermark in a latent-diffusion decoder [32, 37, 33]. RingID extends ring-based signals to multi-key identification, while WIND separates watermark detection into two stages for public use [38, 39]. These methods differ in payload, perceptual objective, and robustness profile, but their common public interface is still usually a detector score or decoded identity.

Provenance requires a more precise interpretation than watermark presence. A positive score means that the submitted image resembles the detector’s expected signal strongly enough under a calibration rule. It does not necessarily bind that signal to one prompt, seed, model, resolution, or committed execution trace. Likewise, signed content credentials authenticate the signed assertions and their chain, but the assertion is only as strong as the evidence placed behind it. The technical question in this chapter is consequently not whether statistical watermarks or signed manifests should be discarded. It is how a public watermark decision can be coupled to a separately checkable relation describing the claimed generation context.

This distinction yields two complementary service levels. A statistical mode can scan large volumes of transformed images and tolerate small signal degradation. A proof-bearing mode can make a stronger statement about an exact published object, but must carry an explicit proof artifact and generally rejects altered pixels. Treating these modes as interchangeable would obscure both their value and their limits.

3.2.2 Forgery Vulnerabilities of Public Detectors

A removal attack begins with a watermarked image and tries to lower its detection score while preserving acceptable image quality. A forgery attack begins with an unmarked or differently marked image and tries to raise the score associated with a

target source. These goals are asymmetric. Robustness against removal does not imply resistance to forgery, because a highly stable but publicly exposed pattern may be easy to implant in unrelated content.

Black-box imprint attacks demonstrate this distinction. From a watermarked reference, the adversary estimates a transferable imprint and optimizes an arbitrary cover image so that a target detector attributes it to the reference source [8]. The attack does not require disclosure of the target scheme’s internal key derivation. Query access, proxy models, and common latent structure can supply a useful optimization signal. A successful attack therefore produces a positive detector outcome without the generation process that the verifier may implicitly associate with that outcome.

The white-box case is stronger. When the detector or a differentiable surrogate is public, an adversary can maximize the verification statistic directly under a pixel perturbation budget. Prior work shows that watermark-based detectors can be evaded or optimized against when their decision rules are exposed, and public knowledge of a watermark’s structure can enable specialized removal strategies [7, 9]. Rate limits may slow a pure query attack, but they do not create cryptographic evidence that a positive score originated from the registered embedder.

A signature-only construction addresses a different object. If metadata is signed, an adversary without the signing key cannot alter it undetectably. If the signing key is available to a malicious producer or compromised operator, however, the attacker can sign a fabricated binding or a clean image. Adding a score blocks the clean-image case only when the attacker cannot search for a binding whose derived pattern happens to correlate with the image. The resulting problem is a claim–evidence mismatch: the public interface says “verified,” but the evidence may show only that some chosen detector statistic crossed a threshold and some metadata was signed.

The threat model used here grants the adversary watermarked images and all pub-

lic artifacts. The adversary may manipulate pixels, query the public score, and use detector gradients. The PP-Mark binding key k_{sec} is not disclosed, and the adversary is assumed unable to violate SHA-256 collision resistance or SP1 computational soundness. A conventional metadata-signing key used in comparison architectures is separate from k_{sec} ; compromising that comparison key does not reveal the PP-Mark binding key. Conversely, compromise of k_{sec} is an explicit failure of the operational assumption and is not hidden by the theorem.

3.2.3 Zero-Knowledge Proofs and Verifiable Computation

Commit-and-open protocols allow a prover to fix a large object before learning which parts a verifier will inspect. A Merkle tree reduces the committed object to one root; an authentication path then proves that an opened leaf belongs to the committed tree without publishing every leaf [40]. If the opening challenge is derived after the commitment and is bound to the submitted image, a prover cannot freely reuse one favorable opening for a different image. The residual risk is explicit: a random opening can miss all positions at which a committed trace cheats. That probability becomes a term in the guarantee rather than an implicit trust assumption.

Zero-knowledge proofs add a second property. A prover can demonstrate that a private witness satisfies a public predicate without revealing the witness itself. PP-Mark realizes this layer with the SP1 zero-knowledge virtual machine, which can prove execution of a Rust program and publish a standalone receipt [41]. The proof program checks key binding, Merkle commitment, image-bound challenge, and deterministic embedding relations. It does not prove the semantic truth of the context fields, and it does not re-run the complete image generator.

Related provenance systems occupy adjacent points in this design space. C2PA standardizes signed content credentials and manifest relationships, whereas PP-Mark

supplies evidence about a watermark embedding relation that can be carried with such metadata [6]. VeriLLM uses commitment and randomized opening to make decentralized inference publicly checkable; PP-Mark applies a related commit–open–verify idea to sampled image-watermark traces and additionally binds the opening challenge to the exact image hash [42]. ZeroMark hides a dataset watermark during ownership verification, and CLUE-Mark studies undetectability from a lattice assumption; those goals are complementary to the unforgeability target studied here [43, 44].

The key design decision is thus not the mere presence of a zero-knowledge proof. The public statement must name the same binding and image used by the score, the private witness must encode the committed trace relation, and the final interface must refuse to collapse a score-positive result into full acceptance. Otherwise, a proof can be valid while remaining irrelevant to the provenance sentence shown to the user.

3.3 PP-Mark

Figure 3.1 presents the complete data flow. During generation, the context and secret key derive a public binding; that binding drives both the latent sign pattern and the trace commitment. The generator publishes metadata, a Merkle root, and a proof receipt alongside the image. During verification, a third party inverts the image to estimate its initial latent, recomputes the score, hashes the canonical pixels, and verifies the receipt. No central detector API or disclosure of the binding key is required.

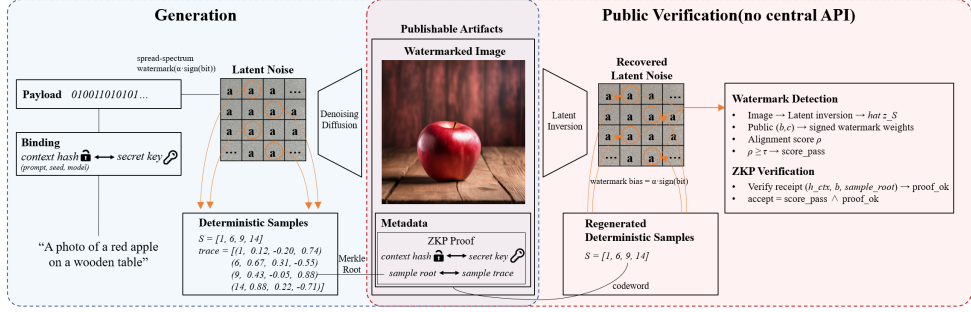


그림 3.1: PP-Mark generation and public verification workflow

3.3.1 Context-Bound Binding and Payload Construction

Let p be the prompt, s the generation seed, m_{id} the model identifier, and $W \times H$ the output resolution. Domain-separated hashing first constructs a context hash and then mixes that public context with the secret binding key:

$$\begin{aligned}
 h_p &= H(p), \\
 h_{ctx} &= H(\text{ctx_hash} \parallel h_p \parallel m_{id} \parallel s \parallel W \parallel H), \\
 b &= H(\text{ctx_hash} \parallel h_{ctx} \parallel k_{sec}).
 \end{aligned} \tag{3.2}$$

Changing the prompt, seed, model identifier, or resolution changes h_{ctx} and hence b . The prompt itself need not be placed in the proof witness; its digest is sufficient for the binding relation. Domain separation prevents the same hash invocation from being interpreted as another protocol field.

The public binding becomes a fixed-length watermark message and codeword:

$$m_{pl} = \text{Trunc}(b), \quad c_{RS} = \text{RS}(m_{pl}), \quad \mathbf{r} = \text{Bits}(c_{RS}). \tag{3.3}$$

Reed–Solomon expansion supplies a redundant binary sequence $\mathbf{r}$ that can be

spread across the latent space. This sequence is public once b is published, but the proof must still show that b is consistent with a valid secret-key witness. Public reconstructability is necessary for independent score computation; it is not permission to choose a new binding after inspecting the target image.

The exact image enters after the trace has been committed. Let root_{tr} denote the Merkle root of the sampled trace and let h_{img} be SHA-256 of deterministically canonicalized pixels. The public challenge material is

$$\begin{aligned} u &= (h_{\text{img}}, h_{\text{ctx}}, \text{root}_{\text{tr}}), \\ \text{seed}_{\text{open}} &= H(\text{ppmark_opening_v1} \parallel u), \\ \mathcal{O} &= \text{DeriveOpen}(\text{seed}_{\text{open}}, k_{\text{open}}, N_{\text{tr}}). \end{aligned} \tag{3.4}$$

This ordering matters. The root fixes the trace before the opening set is known, and h_{img} prevents a receipt prepared for one exact image from being replayed unchanged on another. A cryptographic pixel hash is chosen instead of a perceptual hash because a perceptual collision neighborhood would create an explicit replay region. The consequence is intentional but operationally important: recompression, crop, resize, or any other pixel change invalidates the receipt and requires proof re-issuance for the edited variant.

3.3.2 Deterministic Sampling and Latent Watermark Embedding

The codeword deterministically selects N_{tr} latent locations,

$$\mathcal{S} = \text{DeterministicSample}(c_{\text{RS}}, N_{\text{tr}}), \quad \text{idx}(x, y) = H(b \parallel x \parallel y \parallel 777) \bmod L, \tag{3.5}$$

where $L = |\mathbf{r}|$. Any verifier given the same binding and public parameters therefore reconstructs the same sample positions and coordinate-to-bit mapping. The im-

plementation uses one fixed reference latent channel. Deterministic spreading lets the detector aggregate a weak local signal over many positions and keeps the sampled trace auditable.

For every latent index i , the binding also determines a pseudo-random Gaussian baseline. The construction maps a 32-bit hash-derived integer through a fixed lookup table for the inverse normal CDF, F^{-1} :

$$\begin{aligned} u_i &= H(b \parallel (i + 1)) \bmod 2^{32}, \\ g_i &= F^{-1}\left(\frac{u_i}{2^{32}}\right), \\ r_i &= 2 \mathbf{r}_{\text{idx}(x,y)} - 1, \\ z_i &= g_i + \alpha r_i. \end{aligned} \tag{3.6}$$

Here $r_i \in \{-1, +1\}$ is the watermark sign and α controls embedding strength. To preserve stable latent variance, the initialized latent and effective strength are normalized as

$$z_i^{(0)} = \frac{z_i}{\sqrt{1 + \alpha^2}}, \quad \alpha_{\text{eff}} = \frac{\alpha}{\sqrt{1 + \alpha^2}}. \tag{3.7}$$

For each $i \in \mathcal{S}$, the generator records the tuple $(i, g_i, z_i^{(0)}, \beta_i)$ in Q18 fixed-point form, where the committed bit $\beta_i = \mathbf{r}_{\text{idx}(i)}$ for an honestly generated trace. Those tuples constitute $\mathcal{T}_{\text{tr}}$. Fixed-point representation makes the proof program's arithmetic deterministic; it also means that the verified relation is the registered Q18 relation rather than an unspecified real-number computation.

The public verifier applies DDIM inversion to the candidate image I and obtains an estimate $\hat{z}^{(0)}$. The expected watermark component at a sampled position is $w_i = \alpha_{\text{eff}} r_i$. The normalized correlation score is

$$S_{\text{det}} = \frac{\left| \left\langle \hat{z}_{\mathcal{S}}^{(0)}, w_{\mathcal{S}} \right\rangle \right|}{\|w_{\mathcal{S}}\|_2}. \quad (3.8)$$

The score interpretation depends on inversion error. Write $\epsilon_i = \hat{z}_i^{(0)} - z_i^{(0)}$. The source analysis uses the following empirical statistical condition.

Assumption 3.1 (Inversion-residual condition). *The residuals ϵ_i are approximately zero mean and have bounded moderate correlation with the watermark signs r_i over the registered sample positions.*

Under Assumption 3.1, the watermark term accumulates coherently, while the Gaussian baseline and residual terms do not systematically cancel it. This is not asserted as an architecture-independent fact. It is checked empirically for Stable Diffusion 2.1 and SDXL in Section 3.4.4.

3.3.3 Trace Commitment and SP1 Proof Generation

The sampled Q18 trace is committed with a Merkle tree. A leaf hashes the sample index, Gaussian baseline, normalized latent value, and payload bit; internal nodes hash ordered child values until one root root_{tr} remains. The root fixes the trace economically, while authentication paths let the proof program check membership of opened tuples [40]. The opening digest commits the ordered tuples selected by $\mathcal{O}$.

The SP1 receipt π attests that there exists a private witness

$$\omega_{\text{priv}} = (k_{\text{sec}}, c_{\text{RS}}, \mathcal{T}_{\text{tr}}, \text{paths}) \quad (3.9)$$

consistent with the public statement

$$x_{\text{pub}} = (b, \text{root}_{\text{tr}}, \text{digest}_{\text{open}}, \text{seed}_{\text{open}}, k_{\text{open}}, \alpha_{\text{eff}}, h_{\text{ctx}}, \\ H(F^{-1}), N_{\text{tr}}, h_{\text{img}}). \quad (3.10)$$

The image hash is shown explicitly in Equation (3.10) because it is an input to the opening challenge even though image pixels are not a private circuit witness. The lookup table hash fixes the numerical inverse-CDF implementation, and the public sample count makes the trace dimension auditable.

The proof program evaluates a composite predicate

$$\mathcal{P}(x_{\text{pub}}, \omega_{\text{priv}}) = P_1 \wedge P_2 \wedge P_3 \wedge P_4, \quad (3.11)$$

with the following obligations:

$$P_1 : \quad b = H(\text{ctx_hash} \parallel h_{\text{ctx}} \parallel k_{\text{sec}}), \quad (3.12)$$

$$P_2 : \quad \text{MerkleRoot}(\mathcal{T}_{\text{tr}}) = \text{root}_{\text{tr}}, \quad (3.13)$$

$$P_3 : \quad \begin{aligned} \mathcal{O} &= \text{DeriveOpen}(\text{seed}_{\text{open}}, k_{\text{open}}, N_{\text{tr}}), \\ \text{digest}_{\text{open}} &= H(\{\text{leaf_tuple}_i\}_{i \in \mathcal{O}}), \end{aligned} \quad (3.14)$$

$$P_4 : \quad \forall i \in \mathcal{O} : \quad \begin{cases} \beta_i = \mathbf{r}_{\text{idX}}(i), \\ u_i = H(b \parallel (i + 1)) \bmod 2^{32}, \\ g_i = F^{-1}(u_i / 2^{32}), \\ z_i^{(0)} = g_i + \alpha_{\text{eff}}(2\beta_i - 1) \pmod{\text{Q18}}. \end{cases} \quad (3.15)$$

P_1 binds the public value to possession of the secret-key witness. P_2 fixes the sampled trace under the published root. P_3 binds the opened tuples to the image-derived challenge. P_4 checks that every opened tuple follows the binding-derived payload, Gaussian lookup, and registered embedding equation. The last obligation is essential: merely checking that a committed tuple satisfies an arithmetic relation chosen by the prover would permit tautological witnesses.

The circuit’s negative boundary is as important as its checked obligations. It does not execute the text encoder, denoising trajectory, decoder, canonicalization outside the registered hash input, or full DDIM inversion inside SP1. Image-side signal presence is evaluated by Equation (3.8); the proof side checks consistency of the opened committed relation. Full acceptance joins these two independently visible conditions.

3.3.4 Public Verification Modes and Formal Security Guarantee

PP-Mark exposes two modes rather than one ambiguous “verified” label.

PP-Mark(score). The verifier reconstructs the binding-derived sign pattern, DDIM-inverts the image, computes S_{det} , and reports whether $S_{\text{det}} \geq \tau$. The threshold is calibrated on clean images for a target false-positive rate. This mode is fast and can tolerate moderate lossy transformations through soft correlation and alignment search. It is only statistical screening. It does not establish that a secret-key holder committed a trace or that the image is accompanied by a valid receipt.

Full acceptance. The verifier additionally canonicalizes the pixels, computes h_{img} , reconstructs the image-bound opening challenge, and verifies π against x_{pub} . It returns a positive decision only when both the score and receipt checks pass, as in Equation (3.1). This mode is appropriate when the exact published object and a portable verification artifact are available. Its meaning is “verified relative to the declared binding,” not “factually true” or “human-authored.”

Table 3.1 records the different surfaces covered by the two modes. A score can screen an external attack under the calibrated detector, but a malicious generator that never embedded the registered signal can choose metadata adversarially. The proof-bearing mode adds a check on the binding and committed trace without disclosing

k_{sec} .

⌘ 3.1: Threat coverage by verification mode

Threat condition	Score	Accept
External adversary without k_{sec}	✓	✓
Generator fraud without registered embedding	×	✓
Third-party audit without key disclosure	×	✓

The formal statement makes the remaining failure probability visible. Let $\mathcal{G}_{\text{legit}}$ be the set of image–receipt pairs produced by the registered generator, and suppose a committed trace contains at least q_{bad} positions that violate the honest embedding relation.

Assumption 3.2 (Cryptographic verification condition). *SHA-256 is collision resistant and is modeled as a random oracle for derivation of the post-commitment opening challenge [45]; SP1 is computationally sound for security parameter λ ; and the adversary does not possess k_{sec} .*

Theorem 3.1 (Unforgeability with partial opening). *Under Assumption 3.2, for every probabilistic polynomial-time adversary $\mathcal{A}$ that commits a trace with at least q_{bad} violating positions,*

$$\begin{aligned}
& \Pr[\text{PPMark}_{\text{accept}}(I^*, \text{md}^*, \pi^*) = 1 \wedge (I^*, \pi^*) \notin \mathcal{G}_{\text{legit}}] \\
& \leq \text{Adv}_H^{\text{cr}}(\mathcal{A}) + \text{Adv}_{\text{SP1}}^{\text{sound}}(\mathcal{A}) + \frac{\binom{N_{\text{tr}} - q_{\text{bad}}}{k_{\text{open}}}}{\binom{N_{\text{tr}}}{k_{\text{open}}}} \\
& \leq \text{Adv}_H^{\text{cr}}(\mathcal{A}) + \text{Adv}_{\text{SP1}}^{\text{sound}}(\mathcal{A}) + \left(1 - \frac{q_{\text{bad}}}{N_{\text{tr}}}\right)^{k_{\text{open}}}.
\end{aligned} \tag{3.16}$$

Proof sketch. A verifying forged receipt has three relevant routes. Reusing a public binding with a different secret witness violates the hash-binding relation, and alter-

ing a trace while preserving its root violates Merkle collision resistance. Otherwise, SP1 soundness implies that the composite predicate has a satisfying witness except with its stated advantage. Conditional on fixed commitments, the image-bound challenge selects a uniformly modeled k_{open} -subset of the N_{tr} positions. The receipt can avoid detection of all q_{bad} inconsistent positions only when the opening contains none of them. The exact hypergeometric probability of that event is the binomial ratio in Equation (3.16), which is at most $(1 - q_{\text{bad}}/N_{\text{tr}})^{k_{\text{open}}}$. The union bound over hash, proof-soundness, and missed-opening events yields the result. Requiring the score can only further restrict acceptance; it does not enlarge this upper bound. $\square$

For the implemented $N_{\text{tr}} = 1000$ and $k_{\text{open}} = 32$, the partial-opening term is approximately $2^{-6.3}$ when $q_{\text{bad}} = 0.125N_{\text{tr}}$, corresponding to a 98.7% chance that the opening intersects the cheating set. When half of the trace positions violate the relation, the term is 2^{-32} . Small cheating sets are less likely to be opened, which is why the calibrated score remains a separate condition. This is a two-layer defense under explicit assumptions, not a claim that either layer detects every possible deviation.

The theorem also does not repair key compromise. If k_{sec} is disclosed, an adversary can construct bindings and witnesses under that key. Rotation, epoch identifiers, public revocation information, and secure key storage are therefore deployment obligations. Similarly, if a proof artifact is detached or corrupted, the image returns to an unverified state rather than falling back to a stronger claim based on its score.

3.4 Experimental Evaluation

3.4.1 Experimental Setup and Calibration

The primary experiments use Stable Diffusion 2.1 base with a DDIM scheduler, 50 inference steps, and classifier-free guidance scale 7.5. Prompts are drawn from

the Stable-Diffusion-Prompts collection. Unless an ablation states otherwise, PP-Mark uses embedding strength $\alpha = 4.0$, $N_{\text{tr}} = 1000$, and $k_{\text{open}} = 32$. Generation, embedding, inversion, and detector measurements run on an NVIDIA H100 PCIe 80 GB GPU; SP1 measurements use the same server with an Intel Xeon Gold 6542Y processor.

Each detector threshold targets a 1% false-positive rate (FPR) on clean images generated under the matching model settings. The main forgery metric is forgery acceptance rate (FAR) at that threshold. Removal and transformation experiments report true-positive rate (TPR) after the attack. A score-mode result concerns only the calibrated detector criterion. Full acceptance additionally requires a valid receipt. Keeping those denominators distinct prevents a robust score from being misreported as cryptographic acceptance.

The first experiment isolates the acceptance architecture before comparing watermark methods. Four interfaces are evaluated on 100 images under two targeted attacks. ‘Sig-only’ accepts correctly signed metadata. ‘Sig+Score’ requires a signature and a positive detector score. PP-Mark(score) fixes the binding through the secret-key construction but performs only statistical screening. PP-Mark(accept) adds the SP1 receipt.

In the fabricated-binding attack, an adversary controlling only the comparison metadata-signing key tests 10,000 random candidate bindings per image and retains the one with the highest score. The mean best score is 3.88, above the calibrated $\tau = 2.47$. In the compliance-bypass attack, the adversary signs plausible metadata for a clean image without embedding the registered signal. Table 3.2 reproduces the source outcomes.

⌘ 3.2: Forgery acceptance by verification architecture

Attack	Sig-only	Sig+Score	PP-Mark (score)	PP-Mark (accept)
Fabricated binding	1.00	1.00	$\leq 0.01^\dagger$	0.00
Compliance bypass	1.00	0.00	$\leq 0.01^\dagger$	0.00

FAR at a 1% clean-calibrated FPR; $n = 100$ per attack. † The bound is the calibrated FPR; 0 of 100 trials were observed positive.

Signature-only verification accepts both attacks because it authenticates metadata controlled by the adversary, not the embedding relation. Adding a score blocks a clean submission but does not block search over bindings: all 100 instances find a score-positive candidate. The secret-key-derived PP-Mark binding removes that search freedom, and neither 100-image set produces an observed score positive. Full acceptance also checks the receipt and records zero observed acceptances. These finite results support the conjunctive interface; they do not replace the conditional bound in Theorem 3.1.

A post-submission signature control further separates exact integrity from embedding compliance. Bare-Sig uses Ed25519 over the canonical pixel hash; Context-Sig additionally signs the context hash and b . Both are appropriate exact-image integrity controls under a trusted signer, but neither proves that the signer actually embedded a watermark. Table 3.3 reports the fixed 100-image comparison.

For exact integrity alone, a conventional signature is simpler and preferable. PP-Mark’s additional contribution is different: the receipt and score jointly test an embedding relation, while the image-side signal remains detectable after selected transformations that invalidate exact signatures. The comparison does not claim that a large proof should replace a signature when no embedding- compliance statement is required.

⌘ 3.3: Exact-signature and embedding-compliance controls

Test ($n = 100$)	Bare-Sig	Context-Sig	PP-Mark evidence
Original verifies	100/100	100/100	–
Cross-image replay accepted	0/100	0/100	–
Context substitution accepted	100/100	0/100	–
Authorized signer signs an unwatermarked image	100/100	100/100	0/100 score or accept
JPEG quality 25 after signing	0/100 valid	0/100 valid	94/100 score detections
75% crop–resize after signing	0/100 valid	0/100 valid	87/100 score detections

3.4.2 Forgery Attacks

Black-box imprint transfer. The black-box evaluation tests whether an imprint extracted from one watermarked reference can be transferred to an arbitrary cover image and attributed to the reference source [8]. A fixed pool contains 200 prompts, split equally between calibration and evaluation, and 100 fixed cover–reference pairs are reused for every method. Stable Diffusion 2.1 serves as both the target and proxy architecture. Optimization budgets of 50, 100, and 150 steps are evaluated.

Before cross-method transfer, a direct sanity check attacks Tree-Ring and Gaussian Shading with their own imprints. Tree-Ring reaches FARs from 0.92 to 0.97 across the three step budgets, and Gaussian Shading reaches 1.00 throughout. The pipeline therefore produces the targeted attribution effect before it is applied to other detectors.

Table 3.4 aggregates the three step budgets for Tree-Ring-sourced and Gaussian-Shading-sourced imprints. The source table also reports percentage changes against PP-Mark(score), but those ratios use only one and two PP-Mark positive trials, respectively. Absolute FAR, confidence intervals, and one-sided Fisher tests are retained as

the primary evidence.

⌘ 3.4: Cross-method black-box imprint transfer

Method	FAR	95% CI	Relative Δ (%)	Fisher p
<i>Tree-Ring-sourced imprint</i>				
TrustMark	0.0367	0.0184–0.0647	+1000.0	0.0029
InvisMark	0.0300	0.0133–0.0500	+800.0	0.0102
RingID	0.0400	0.0208–0.0688	+1100.0	0.0016
StableSig	0.0333	0.0161–0.0604	+900.0	0.0055
HiDDeN	0.0233	0.0094–0.0475	+600.0	0.0343
WIND	0.0100	0.0021–0.0289	+200.0	0.3119
PP-Mark(score)	0.0033	0.0001–0.0184	–	–
PP-Mark(accept)	0.0000	–	–	–
<i>Gaussian-Shading-sourced imprint</i>				
TrustMark	0.0267	0.0116–0.0519	+300.0	0.0532
InvisMark	0.0167	0.0033–0.0333	+150.0	0.2252
RingID	0.0133	0.0036–0.0338	+100.0	0.3430
StableSig	0.0167	0.0054–0.0385	+150.0	0.2252
HiDDeN	0.0233	0.0094–0.0475	+250.0	0.0882
WIND	0.0100	0.0021–0.0289	+50.0	0.5000
PP-Mark(score)	0.0067	0.0008–0.0239	–	–
PP-Mark(accept)	0.0000	–	–	–

FAR at 1% FPR, aggregated across 50, 100, and 150 optimization steps. Relative Δ is descriptive because its PP-Mark(score) denominators are one positive Tree-Ring-sourced trial and two positive Gaussian-Shading-sourced trials.

PP-Mark(score) remains at or below the 1% calibration floor: 0.0033 for the Tree-Ring source and 0.0067 for the Gaussian-Shading source. PP-Mark(accept) records no accepted forgeries. Tree-Ring-sourced differences for RingID, Stable Signature, and HiDDeN are significant at 0.05 in the reported Fisher tests; Gaussian-Shading-sourced comparisons do not reach that level. The latter result is not converted into a claim of

method equality. The absolute ordering and intervals show low PP-Mark transfer in this fixed experiment, while the full mode additionally refuses a receipt-free attribution.

White-box optimization. The white-box attacker receives each detector and its gradients and applies 150 projected-gradient steps under an ℓ_∞ budget of $2/255$. A score-mode forgery succeeds when the method’s calibrated detector crosses its threshold. Average step counts and peak signal-to-noise ratio (PSNR) are computed only among successful forgeries. Table 3.5 reports the Stable Diffusion 2.1 outcomes.

⌘ 3.5: White-box forgery outcomes on Stable Diffusion 2.1

Method	FAR at 1% FPR	Avg. steps to FAR	Avg. PSNR
TrustMark	1.00	2.41	75.73
InvisMark	1.00	3.07	72.67
StableSig	1.00	5.40	65.91
HiDDeN	1.00	8.08	62.96
RingID	1.00	2.26	74.51
WIND	1.00	22.20	55.76
PP-Mark(score)	0.76	17.42	67.69
PP-Mark(accept)	0.00	–	–

Every comparison detector reaches a 1.00 FAR in this white-box setting. PP-Mark(score) is also vulnerable to direct optimization, reaching 0.76 FAR. This result is precisely why a score-positive image is not labeled as publicly verified provenance. Pixel optimization changes h_{img} , hence the opening seed, opened positions, and opening digest. The attacker does not produce a matching receipt, so PP-Mark(accept) records zero full acceptances. The observed zero is conditional on the tested budget

and sample and is interpreted together with the cryptographic assumptions, not as an attack-independent empirical probability.

3.4.3 Removal and Transformation Attacks

Forgery resistance is not sufficient if ordinary manipulation always destroys the screening signal. The next experiments therefore evaluate score-mode retention under regeneration, multi-image steganalysis, gradient-free removal, and common pixel or geometric transformations. These tests do not claim that an old exact receipt remains valid after editing; by design, any changed canonical pixel string requires a new receipt.

Regeneration. Diffusion regeneration adds noise to a watermarked image and synthesizes it again, a process known to remove some invisible pixel watermarks [36]. Table 3.6 reports detector TPR at three source noise-step settings. PP-Mark retains 0.98 in all three, while several post-hoc methods fall below 0.10. Tree-Ring also remains high at 0.93–0.95, consistent with the relative durability of signals embedded in the generation latent.

⌘ 3.6: Detection after diffusion regeneration

Method	30 steps	60 steps	100 steps
StableSig	0.04	0.02	0.03
HiDDeN	0.10	0.06	0.04
TrustMark	0.85	0.63	0.39
InvisMark	0.01	0.05	0.02
Tree-Ring	0.95	0.93	0.93
PP-Mark	0.98	0.98	0.98

Values are score-mode TPR at the 1% clean-calibrated FPR.

Steganalysis. The multi-image attack averages residuals from K watermarked reference images, treats the average as a shared watermark estimate, and subtracts it from a target. Table 3.7 shows that Stable Signature and Tree-Ring lose much of their detection rate as more references are averaged. TrustMark, InvisMark, WIND, and PP-Mark remain highly detectable at larger K , consistent with image-conditioned patterns that do not converge to one reusable residual.

⌘ 3.7: Detection after multi-image steganalysis

Method	$K = 10$	$K = 100$	$K = 500$	$K = 1000$
StableSig	0.200	0.200	0.173	0.171
Tree-Ring	0.190	0.020	0.038	0.035
TrustMark	0.990	0.960	0.960	0.960
InvisMark	1.000	1.000	1.000	1.000
WIND	1.000	1.000	1.000	1.000
PP-Mark	0.864	0.959	0.963	0.958

Values are score-mode TPR at the 1% clean-calibrated FPR.

Gradient-free SPSA. Simultaneous perturbation stochastic approximation (SPSA) estimates a search direction with score queries rather than detector gradients [46]. For removal, the adversary minimizes the PP-Mark score for 50 steps. As shown in Figure 3.2, the score can cross below τ , but only after image quality collapses: successful low-score points have PSNR from 4.7 to 6.9, and no removal satisfies the source validity criterion of PSNR at least 30. This result is evidence under one attack and quality constraint, not a general impossibility of black-box removal.

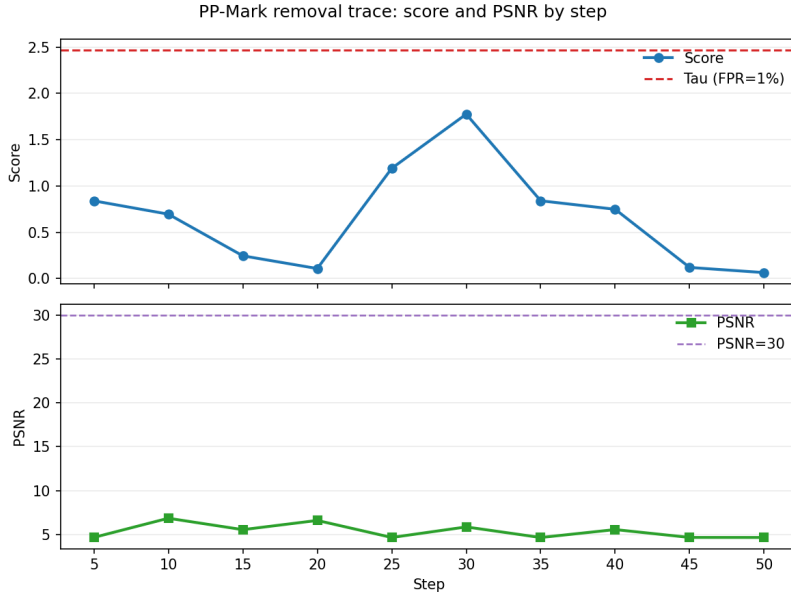


그림 3.2: SPSA removal trajectory

In the complementary SPSA forgery test, score-mode FARs at 50, 100, and 150 steps are 0.02, 0.00, and 0.01, with corresponding mean PSNRs 6.55, 6.61, and 6.63. The aggregate FAR is 0.01 at mean PSNR 6.60. The low-quality solutions again show that reporting attack success without a perceptual constraint would be misleading. Full acceptance would additionally require a receipt for the changed image.

Pixel and geometric transformations. The detector uses an alignment search for geometric synchronization. Table 3.8 reports score-mode TPR for Stable Diffusion 2.1 and SDXL. Compression, blur, additive noise, and brightness changes retain TPRs from 0.86 to 1.00. Random crop with resize is the weakest SDXL case at 0.74, while a 75-degree rotation reaches 0.85 on SD 2.1 and 0.97 on SDXL after synchronization.

⌘ 3.8: Detection after pixel and geometric transformations

Transformation	SD 2.1 TPR	SDXL TPR
JPEG, quality 25	0.94	1.00
Blur, 8×8	1.00	1.00
Gaussian noise, $\sigma = 0.1$	0.86	0.95
Brightness, $\times 0.6$	0.98	1.00
Random crop and resize, 75%	0.87	0.74
Rotation, 75°	0.85	0.97

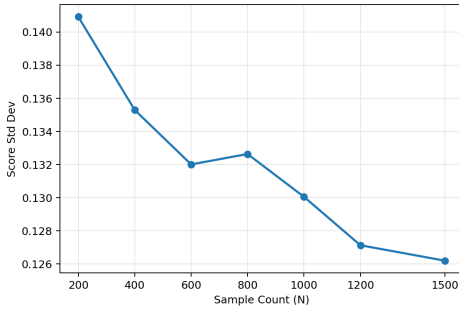
Score-mode TPR at the matching 1% FPR; SDXL crop-and-resize uses $n = 50$.

These transformation results define the role of PP-Mark(score), not the validity of an old proof. The canonical SHA-256 image hash changes under every listed operation. A transformed image may still be discovered by the statistical screen, but it becomes PP-Mark-accepted only after a producer issues a receipt for that exact transformed pixel representation.

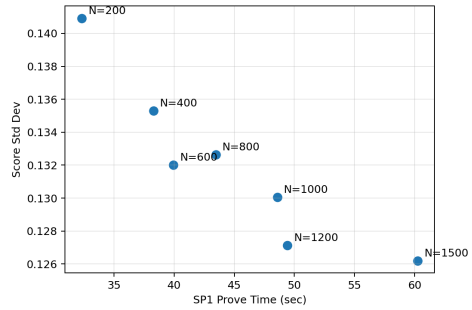
A post-submission sequential test applies 75% crop, bicubic resize, and JPEG quality 25 to the same 100 SD 2.1 watermarked images with no survivor filtering. Persistent success requires detection at every checkpoint; 80 of 100 images remain detectable through the complete pipeline, compared with 87 of 100 after crop–resize alone and 94 of 100 after JPEG alone. The 80% result measures cumulative score degradation under one fixed three-operation pipeline. It does not change the exact-image boundary of PP-Mark(accept).

3.4.4 Efficiency, Image Quality, and Model Generalization

The sample count controls detector variance, proof work, and receipt size. The two source ablations appear in Figure 3.3. Increasing the count from 200 to 1000 reduces the empirical score standard deviation from 0.1409 to 0.1301, a 7.7% decrease; increasing it from 1000 to 1500 supplies only a further 3.0% decrease. SP1 proving time grows from 32.309 s at 200 samples to 48.572 s at 1000 and 60.272 s at 1500. The selected value of 1000 is therefore an empirical operating point rather than a universal optimum. Concentration with larger samples supplies the statistical motivation, while the source analysis uses the registered empirical calibration rather than claiming an asymptotic detector guarantee [47].



(a) Score variation



(b) Proving time

그림 3.3: Sampling stability and SP1 proving trade-off

Table 3.9 reports end-to-end runtimes on Stable Diffusion 2.1. The score mode matches the Tree-Ring path at 0.89 s for embedding and 0.72 s for verification. Full acceptance adds 48.6 s for one-time proof generation and 7.59 s for receipt verification. At $N_{tr} = 1000$, the more precise SP1 checkpoints are 48.572 s proving and 7.589 s verification, and the receipt occupies 40.68 MB. The artifact is therefore feasible for archival or on-request verification in the evaluated environment, but it is costly for a

per-upload streaming path.

Table 3.9: End-to-end runtime on Stable Diffusion 2.1

Method	Embed or generate (s)	Verify (s)
Tree-Ring	0.89	0.75
Gaussian Shading	3.10	0.72
RingID	0.89	0.72
WIND	0.89	1.10
Stable Signature	0.91	0.008
HiDDeN	0.001	0.001
TrustMark	0.013	0.004
InvisMark	0.002	0.009
PP-Mark(score)	0.89	0.72
PP-Mark(accept)	$0.89 + 48.6^\dagger$	$0.72 + 7.59^\dagger$

[†] Additional SP1 proving and receipt-verification time at $N_{\text{tr}} = 1000$.

After the submitted-paper evaluation, an SP1 v6.3.1 port with Merkle-work optimization was rerun at the same $N_{\text{tr}} = 1000$ configuration. It reduced measured proving time from 48.572 s to 17.0 s and the receipt artifact from 40.68 MB to 2.81 MB. This is reported as a post-submission implementation update, not silently substituted into Table 3.9 or the submitted attack comparisons. The update improves the engineering point but leaves the assurance predicate and the distinction between routine score screening and one-time exact- image proof generation unchanged.

Receipt size scales from 12.24 MB at 300 sampled positions to 65.96 MB at 1500; the private witness grows from 254.69 KB to 1493.57 KB over the same range, while the public input remains approximately 0.66 KB. A STARK-to-SNARK wrapper or

proof aggregation could reduce distributed proof size, but that optimization is not implemented or counted as a current result.

Embedding strength is selected with a tail-sensitive perceptual diagnostic. Figure 3.4 plots the 95th percentile of BRISQUE against α . Values from 3.0 through 4.0 remain approximately 34.1–34.9, whereas 4.5 rises to about 40.5. The implemented $\alpha = 4.0$ retains the stronger detection setting before this observed tail-quality degradation. The underlying study also reports KID 0.0011014 on one 500-image, 20-split configuration; because that is one moderate-size setup, it is treated as supporting distributional evidence rather than a definitive perceptual guarantee.

Post-submission matched-pool measurements place this result beside the evaluated baselines in Table 3.10. Lower BRISQUE is preferred, KID is interpreted toward zero, and higher CLIP alignment is preferred. PP-Mark does not have the best value in every column; the defensible conclusion is that its quality is comparable to several latent- and pixel-watermark baselines while its full mode adds a different embedding-compliance guarantee.

표 3.10: Post-submission matched image-quality comparison

Method	BRISQUE p95 ↓	KID $\times 10^3$ ($\rightarrow 0$)	CLIP $\times 100$ ↑
PP-Mark	34.14	1.101 ± 0.202	31.64
Gaussian Shading	27.35	0.621 ± 1.096	32.42
HiDDeN	105.86	7.239 ± 1.474	31.30
InvisMark	31.36	-0.900 ± 0.895	32.53
RingID	34.64	-0.989 ± 0.701	32.41
Stable Signature	32.69	-0.814 ± 0.878	32.50
Tree-Ring	41.90	-0.090 ± 0.922	32.30
WIND	27.42	-0.386 ± 0.879	32.28
TrustMark	29.44	-0.174 ± 1.001	32.54

KID is an unbiased finite-sample estimator and can be slightly negative; such values are sampling estimates near zero, not negative distances.

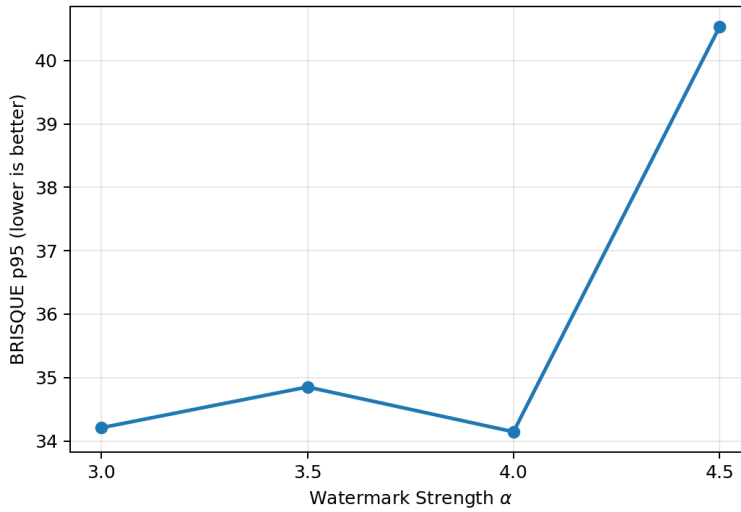


그림 3.4: Perceptual quality across embedding strengths

Inversion diagnostics. Assumption 3.1 is checked on 100 watermarked images per latent-diffusion model. On Stable Diffusion 2.1, the residual mean is 0.011 ± 0.003 , the standard deviation is approximately 0.50, and the mean, median, and maximum absolute residual–sign correlations are 0.30, 0.30, and 0.48. The observed clean watermark score has mean 25.7 and minimum 17.4. On SDXL, the residual mean is -0.022 ± 0.004 , with 95% interval $[-0.031, -0.013]$, standard deviation approximately 0.44, and mean, median, and maximum absolute correlations 0.54, 0.55, and 0.66. Correlation is stronger than on SD 2.1, but it acts as shrinkage rather than complete cancellation in the reported sample.

Bounded SDXL replication. The SDXL experiment uses 100 clean and 100 watermarked images per method, 50 DDIM steps, $\alpha = 4.0$, and $N_{\text{tr}} = 1000$ on the 128-by-128 latent. All thresholds are recalibrated to 1% FPR on SDXL clean images. PP-Mark’s mean watermarked score is 13.63 against $\tau = 2.47$, a 5.5-fold margin, and all 100 clean watermarked evaluations cross the threshold.

Under Tree-Ring-sourced transfer, score-mode FARs at 50, 100, and 150 steps are 0.02, 0.00, and 0.01; under Gaussian-Shading-sourced transfer they are 0.01 at every step. The full mode records zero acceptances in all six conditions. In the SDXL white-box test, TrustMark and InvisMark each reach 1.00 FAR. PP-Mark’s score mode reaches 0.11 with 95% confidence interval 0.0562–0.1883, while the full mode records 0.00. These results show that the registered construction can be run on two evaluated latent-diffusion models. They do not establish architecture-agnostic generalization.

Limitations. Four limitations delimit the experimental conclusion. First, proving time and receipt size remain substantial at the selected sample count. Second, the exact image hash prevents unchanged receipt reuse after a benign edit; re-proving costs about 48.6 s in the submitted environment and 17.0 s in the post-submission

SP1 v6.3.1 update. Third, the receipt must remain attached or retrievable, and key rotation and revocation must operate correctly. Fourth, geometric alignment remains imperfect for crop-and-resize on SDXL, and only SD 2.1 and SDXL are tested. The interface also does not account for cumulative information exposed by repeated adaptive score queries. That dimension remains future work rather than a completed result in this chapter.

3.5 Summary

This chapter developed the Secure Public Verification axis of Claim-Aligned Assurance. A hidden detector limits independent audit, whereas a public detector exposes a score that can be queried or optimized. PP-Mark therefore keeps statistical screening separate from full acceptance, which requires both image-side evidence and a publicly verifiable receipt.

The prompt hash, seed, model identifier, resolution, and producer-held key derive a binding that determines the payload, sampled positions, Gaussian baselines, and latent signs. Q18 trace tuples are Merkle-committed, and the exact image hash derives the post-commitment opening. SP1 checks key binding, trace commitment, the image-bound opening, and the opened embedding relation without disclosing the secret key.

The two evidence layers remain experimentally distinct. Search over 10,000 bindings defeats signature plus score, and white-box optimization raises the public score on 76% of the SD 2.1 covers, yet zero full acceptances are observed in the reported forgery tests. Score mode retains 0.74–1.00 TPR over the common transformations and 0.98 after regeneration. The exact-image proof is not transform robust; an edited variant requires a new receipt.

The assurance statement remains conditional on hash resistance, SP1 soundness,

key control, the composite predicate, inversion behavior, and partial opening. The circuit does not prove complete diffusion execution, semantic truth, authorship, or legal identity. Proof cost and persistence, key lifecycle, geometric synchronization, two-model coverage, and adaptive queries remain open limits.

Within those boundaries, PP-Mark demonstrates the chapter's central principle: a public provenance claim should be released only when the visible detector outcome and the registered verification evidence refer to the same context, signal, image, and predicate. Chapter 4 will combine this result with the post-execution privacy audit to state the dissertation-wide principles of Claim-Aligned Assurance.

제 4 장 Conclusion

4.1 Summary

Trust claims about artificial intelligence systems are often communicated through compact labels, parameters, or acceptance decisions.

These outputs can hide the premises that give a claim its meaning. A differential privacy parameter is incomplete without its protected unit, adjacency, transformation, mechanism, and accountant; a watermark score is not by itself evidence that embedding occurred under a declared generation context. This dissertation addressed the gap between such public statements and the mechanisms and evidence that warrant them.

The resulting perspective is Claim-Aligned Assurance. A positive statement is released only when its target, operative mechanism, verification interface, and evidence refer to the same protected or verified object. An incomplete chain yields a narrower qualified statement or an explicit block. The framework is instantiated in two technically distinct settings: evidence-bound auditing of a completed privacy execution and proof-bound public verification of generative-image provenance.

Chapter 2 treated a reported privacy sentence as a query over frozen evidence rather than as a synonym for an accountant output. The validator reopens the raw-to-generated mapping, checks pipeline and runtime relations, recomputes the registered accountant, resolves the requested unit and adjacency, and releases only the exact native or converted wording licensed by the result. The complete contract returned the expected tuple for all 100 allowed and 100 paired non-allow cases. Fixed necessity witnesses and preserved development failures exposed why mapping, stability, runtime, vacuity, and exact wording cannot be collapsed into one scalar check. WISDM

and UCI HAR executions demonstrated query-dependent authorization, while Sepsis and Backblaze rescans showed that window policy can substantially change raw-to-generated influence without itself constituting a DP certificate.

Chapter 3 changed the assurance object to generative image provenance. PP-Mark binds a latent watermark to generation context, commits a sampled embedding trace, and verifies an image-bound opening through an SP1 receipt. PP-Mark(score) provides transformation tolerant statistical screening; PP-Mark(accept) requires both the calibrated score and a valid receipt for the same context and exact image. A 10,000-binding search and white-box optimization exposed score-only acceptance, whereas the evaluated forgery conditions produced no full acceptances. Score-mode true positive rates were 0.74–1.00 over the reported individual transformations and 0.98 after regeneration. These are observed results under the evaluated protocols, not a universal zero forgery claim.

The two studies are not one algorithm. The privacy audit limits what may be said about an existing execution, whereas PP-Mark changes the evidence required by a public provenance predicate. Their common result is that a defensible claim must name its object, bind the operative mechanism, expose an interface with fixed semantics, and carry the evidence required by those semantics.

4.2 Integrated Discussion and Contributions

4.2.1 Auditable Privacy: Evidence-Relative Post-Execution Audit

The privacy contribution begins from a practical condition: training has already finished and the available evidence is fixed. Different stakeholders may still ask whether the same model supports a generated-window, raw-event, patient, owner, or device-level sentence. Those requests are different mathematical claims even when they dis-

play the same base epsilon. The query must therefore be a scientific input to the audit rather than a label selected after one evidence-only verdict.

The validator connects the requested sentence through a chain of raw adjacency, stable transformation, executed mechanism, accountant, and exact wording. It separates the derivation path, public release state, assurance authority, numerical consequence, and issue set. This distinction permits one frozen execution to license a native claim, license a weaker converted claim, and block a stronger claim. A blocked result does not prove that the stronger privacy property is false; it states that the supplied evidence and registered procedure do not license that sentence.

The experimental contribution is correspondingly about decision validity and authority. The 200-case oracle tests controlled conformance rather than estimating error prevalence. WISDM supplies a complete three-query execution; UCI HAR shows that missing raw support and owner bounds correctly stop the conclusion at the published-window unit. Sepsis and Backblaze expose mapping differences at medical and operational scale, while intentionally withholding privacy wording because no executed mechanism and accountant are attached. Separate verifiers establish software and artifact separation from the production path, not execution attestation.

This produces a concrete reuse boundary. If an existing model's frozen evidence licenses the requested unit, the model can be reused with the exact supported wording. If a registered conversion produces a nonvacuous pair, only that weaker pair may be stated. If the requested patient or owner claim is blocked, the base window pair cannot be relabeled; a mechanism and execution native to the stronger unit may be required. Determining the cost and utility of that alternative is a future experiment, not a result silently inferred from the audit.

4.2.2 Secure Public Verification

The provenance study exposed a tension between independent verification and attack surface. A private detector requires trust in its operator; a public detector enables independent screening but exposes a queryable or differentiable decision surface. PP-Mark therefore does not claim to make the score invulnerable. It changes the stronger acceptance rule so that optimizing the public statistic is insufficient without a valid context-, image-, and trace-bound receipt.

The score and receipt establish different facts. The score supplies image-side evidence that can survive selected lossy transformations. The receipt supplies process-side evidence for the registered binding, trace commitment, sampled embedding relation, and exact-image predicate. Full acceptance requires both, allowing third-party verification without a private API or disclosure of the producer secret. This construction is not a replacement for a conventional signature when exact integrity alone is required; its added value is verification of the embedding relation together with a durable in-image signal.

The interface deliberately has three outcomes rather than an implicit fallback. An exact image with a valid receipt can receive PP-Mark(accept). A transformed copy may retain PP-Mark(score) but does not inherit exact-image acceptance. If the public binding or receipt cannot be resolved, the result is unverified rather than negative evidence that an image is not AI-generated. PP-Mark is therefore a positive provenance system for participating pipelines, not a universal synthetic- image classifier.

4.2.3 Claim-Aligned Assurance Design Principles

The integrated analysis supports four dissertation-wide principles.

1. **Claim specification.** A positive statement identifies its target and semantics: the

privacy unit, adjacency, and wording, or the exact image, generation context, and verification mode.

2. **Mechanism binding.** The statement is connected to the mechanism that gives it meaning. Privacy binds transformation, sampling, clipping, noise, accounting, and runtime evidence; provenance binds payload, latent embedding, trace commitment, image-bound challenge, and proof predicate.
3. **Evidence-bound decision.** An output does not inherit the authority of missing evidence. Positive privacy wording requires its mapping, runtime, and accounting relations; full provenance acceptance requires both image-side detection and process-side receipt verification.
4. **Fail-closed assurance.** A missing, inconsistent, invalid, unregistered, or vacuous premise blocks the stronger interpretation. Blocking does not prove the broader property false; it states that the requested conclusion is not licensed by the declared procedure.

The verification interface operationalizes all four principles. Query separation prevents a window result from becoming a patient answer, while separate score and acceptance modes prevent statistical detection from becoming process-bound provenance. The reusable design question is consequently narrow: which observed or proven relations license the exact positive sentence being released?

4.3 Limitations, Broader Impact, and Future Work

4.3.1 Limitations and Assurance Boundaries

The dissertation covers two assurance domains and two principal systems. It does not establish a complete framework for all trustworthy-AI claims. Fairness, safety,

factuality, utility, and governance can require different objects and evidence. The four principles are a design abstraction grounded in the evaluated settings, not a universal certification standard.

The privacy result remains evidence-relative and finite. It assumes a fallible but nonmalicious producer, registered transformations and accountant adapters, and faithful disclosure of the observable bundle. A malicious producer can fabricate a mutually consistent bundle after executing another procedure. The validator cannot distinguish those worlds without authenticated observation, and therefore does not attest execution, prove arbitrary program semantics, or replace the underlying DP proof. The controlled oracle establishes exact behavior on its authored 200 cases but does not estimate how often published systems contain such errors. Sepsis and Backblaze are mapping diagnostics only, not trained DP models or patient-level certificates.

PP-Mark is conditional on hash resistance, proof-system soundness, producer-key control, inversion behavior, the composite predicate, and the partial-opening bound. It is evaluated on Stable Diffusion 2.1 and SDXL, two latent-diffusion models that share a compatible embedding and inversion interface. The submitted SP1 v5 implementation requires about 48.6 seconds and produces a 40.68 MB receipt at 1,000 sampled positions. A post-submission SP1 v6.3.1 port reduced the measured values to 17.0 seconds and 2.81 MB, but this update does not eliminate proof cost or establish a universal deployment profile. Exact-image acceptance is invalidated by pixel edits, the public binding and receipt must remain retrievable, and key rotation, revocation, geometric synchronization, and broader model families remain open.

Claim alignment can prevent a system from saying more than its evidence supports, but it cannot repair false source evidence, a compromised key, or facts outside a proof circuit. It narrows the statement and makes the stopping boundary visible; it does not eliminate every trust assumption.

4.3.2 Broader Impact

Auditable Privacy applies to sequential learning with medical sensor data, clinical records, activity data, and operational logs. Several privacy units can be plausible in each setting.

Explicit units, transformations, runtime relations, and evidence can help developers and auditors avoid presenting generated record accounting as a stronger patient-, user-, or device-level promise. It can also clarify whether an existing model is reusable under narrower wording or whether a new native unit execution is required. This remains an audit of claim support, not clinical certification.

Secure Public Verification can support creators, platforms, archives, and reviewers that verify declared provenance without one private verification API. Its separated modes clarify that a transformed image can retain a detectable signal while requiring a new receipt for exact acceptance. Provenance evidence nevertheless does not prove factual accuracy, human authorship, social legitimacy, or legal ownership.

Fail-closed systems also impose costs. Unavailable evidence may withhold a legitimate claim, proof expense may exclude producers, and key ownership may centralize control. Deployment therefore requires accessible verification, transparent key governance, clear communication of unverified outcomes, and an appeal or re-issuance process.

4.3.3 Future Experimental Validation

The first future study will quantify the boundary between reuse through post-hoc conversion and retraining with native patient-level DP. The study will use the 5,000-patient PhysioNet/CinC 2019 Sepsis subset already audited in Chapter 2, containing 370 sepsis-positive and 4,630 control patients. A source- and label-stratified patient split will allocate 3,500 patients to training (259 positive, 3,241 control), 750 to vali-

dation (55 positive, 695 control), and 750 to testing (56 positive, 694 control), with no patient crossing a split.

Three training conditions will be compared on the same patient partitions: a non-private reference, window-level DP-SGD followed by the evidence-bound audit, and native patient-level DP-SGD. The native condition will sample patients, aggregate each selected patient’s window gradients, clip once per patient, add Gaussian noise, and account at patient adjacency. Target patient-level epsilon values of 2, 4, and 8, public window-contribution caps of 1, 4, 8, and 16, and five fixed seeds will define the comparison grid. Evaluation will report AUPRC, calibration and utility, the licensed or blocked post-hoc privacy statement, membership-inference exposure, training time, and peak memory. The intended output is a measured rule for when an existing window-private model remains reusable and when a native patient-private execution is necessary.

The second future study will make PP-Mark proof artifacts discoverable after distribution. A C2PA-compatible external manifest repository and resolver will store or reference the public binding b , PP-Mark public statement, SP1 receipt, producer signature, model and inversion configuration, and revocation metadata. C2PA is used here as a signed transport and discovery layer; the design does not assume that C2PA provides one universal central registry or proves the PP-Mark embedding relation.

An exact image will first use a manifest identifier, URI, or hard binding to retrieve its proof package. When embedded metadata has been stripped, a soft fingerprint resolver will return candidate manifests for cryptographic and statistical verification rather than directly declaring provenance. Experiments will include metadata stripping, JPEG compression, crop-resize, manifest substitution, cross-image receipt replay, missing records, and key revocation. Metrics will include retrieval Recall@ k , false-acceptance and true-positive rates, p95 lookup and verification latency, registry

storage, and stale-record rejection. The decision policy remains explicit: an exact image with a valid score and proof receives **Accept**; a transformed image may receive **Score**; and an unresolved artifact returns **Unverified**.

Both future studies close concrete gaps exposed by the completed work. The first tests the cost of moving from an audited converted claim to a native protected unit. The second tests how proof-bound public verification survives real distribution when metadata and sidecars may be removed. Neither result is claimed as completed in this dissertation.

Within the scope evaluated here, the dissertation reaches a narrower conclusion. Privacy and provenance claims become defensible not when a system exposes the largest number of metrics, but when each positive statement is limited to the object, mechanism, interface, and evidence that jointly support it. That is the role of Claim-Aligned Assurance in trustworthy AI.

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ABSTRACT

Privacy parameters and watermark decisions do not establish assurance when their targets, mechanisms, interfaces, and evidence are misaligned. This dissertation develops Claim-Aligned Assurance through Auditable Privacy and Secure Public Verification.

The Auditable Privacy study takes frozen evidence from a completed windowed DP-SGD execution together with requested wording and decides which privacy-unit statement the execution can license. It retains an aligned native claim, applies a registered conversion or group consequence when its premises are complete, and blocks positive wording when evidence is missing or the numerical consequence is vacuous. The complete validator matched the expected tuple on 100 allowed and 100 paired non-allow cases. Bound WISDM and UCI HAR executions demonstrated query-dependent authorization, while mapping diagnostics over 5,000 Sepsis patients and more than 27.7 million Backblaze drive-day records exposed how window policy changes raw-to-generated influence.

PP-Mark combines a context-bound latent watermark, trace commitment, and a zero-knowledge proof receipt. Full acceptance requires a signal and receipt for the same context and image. A 10,000-binding search and white-box optimization exposed score attacks, but no full acceptance was observed; transformation true-positive rates were 0.74–1.00 in score mode.

Across the two studies, the dissertation derives claim specification, mechanism binding, evidence-bound decision, and fail-closed assurance. It limits positive privacy and provenance statements to the objects, mechanisms, interfaces, and evidence that

jointly support them.

keywords: Claim-Aligned Assurance, Auditable Privacy, Differential Privacy, DP-SGD, Secure Public Verification, Generative-AI Provenance, Zero-Knowledge Proof